SAINT MICHAEL COLLEGE OF CARAGA

PREDICTING STUDENT ACADEMIC STANDING USING MACHINE LEARNING MODELS

WITH EXPLAINABLE AI

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

By

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CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"**PREDICTING STUDENT ACADEMIC STANDING USING MACHINE LEARNING MODELS WITH EXPLAINABLE AI**"** presented and submitted by **Janryl S. Signar, Diether F. Flores, Trixia Lee C. Lope, Kevin Jun A. Maestro** in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science** is hereby accepted and recommended for Oral Examination.

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DEDICATION

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ABSTRACT

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I. OBJECTIVES

This study aimed to develop the Student Academic Standing Prediction System using machine learning models integrated with Explainable Artificial Intelligence (XAI). Specifically, it sought to identify system requirements, design and develop the system, train and test machine learning models, evaluate model performance, and assess the overall system quality using selected ISO 25010 criteria.

II. METHODOLOGY

This study utilized a **quantitative developmental research design**. Student data were collected through a Google Forms questionnaire and processed for machine learning model training and testing. The system implemented Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes algorithms. These models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Additionally, the developed system was evaluated using selected ISO 25010 criteria, specifically functional suitability, performance efficiency, usability, and reliability.

III. FINDINGS

The comparative evaluation showed that Random Forest achieved the best performance among the five classifiers, with 97.03% accuracy, 97.00% precision, 97.00% recall, 97.00% F1-score, and 99.00% ROC-AUC. The Explainable AI analysis identified GPA Year 3 Semester 1, weekly study hours, GPA Year 2 Semester 2, LMS login frequency, academic satisfaction, and time management as key predictors of academic standing. The ISO 25010 evaluation obtained a Grand Mean of 3.92, interpreted as “Acceptable,” indicating that the system was generally acceptable in terms of functional suitability, performance efficiency, usability, and reliability.

IV. RECOMMENDATION

Future researchers may expand the dataset by including additional student batches, programs, and academic years to improve model generalizability and strengthen the reliability of future prediction results.

The system may be enhanced by integrating official institutional records, subject to data privacy approval, to reduce reliance on self-reported responses and improve data accuracy.

Further testing is recommended using larger datasets and different deployment environments to evaluate the scalability, stability, and performance of the system under more realistic institutional use.

KEYWORDS: student academic standing, predictive analytics, machine learning models, Explainable Artificial Intelligence, ISO 25010.

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INTRODUCTION

1.1 Project Context

In the rapidly advancing field of educational research, Educational Data Mining (EDM) and Learning Analytics (LA) have become important approaches for monitoring student performance, analyzing learning behavior, and supporting institutional decision-making. Traditional assessment methods, such as grades and instructor observations, may not fully capture the different academic, behavioral, and contextual factors that influence student success. EDM and LA address this limitation by using educational data to identify patterns, predict academic outcomes, and support timely interventions [5], [22], [24], [37].

Machine learning algorithms such as Random Forest (RF), Support Vector Machine (SVM), Decision Tree (DT), K-Nearest Neighbors (KNN), and Naive Bayes are commonly applied in student performance prediction because they can classify students based on academic, behavioral, and demographic indicators. Studies show that tree-based and ensemble methods, particularly Random Forest, are effective in handling complex educational datasets and improving predictive performance [4], [7], [15], [21], [31]. These findings support the need to compare multiple classifiers before selecting the most appropriate model for a specific institutional dataset.

Behavioral data can also improve the reliability of academic prediction systems. Indicators such as Learning Management System (LMS) activity, click data, study behavior, and student engagement provide additional context beyond grades alone. Prior studies have shown that combining academic records with behavioral or activity-based data can improve the early identification of students who may require academic support [6], [8], [30], [35]. This is relevant to the present study because the proposed system considers both self-reported academic history and engagement-related indicators, such as study hours, LMS login frequency, and library visits.

Localized predictive analytics is also important because educational institutions differ in available data, student characteristics, academic policies, and resource capacity. Predictive models developed for one institution may not automatically generalize to another setting. For this reason, local datasets and institutional context must be considered when designing student academic standing prediction systems. Studies on educational data mining, academic performance prediction, and learning analytics emphasize the importance of adapting predictive models to the available data and the actual decision-making needs of the institution [3], [6], [9], [20], [28].

This study advances prior work by developing a predictive system that compares five machine learning classifiers: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. The system also integrates Explainable Artificial Intelligence (XAI) to support interpretability and transparency. XAI techniques such as SHAP and LIME are used in machine learning to explain how features contribute to model predictions, helping users understand why a prediction was generated [39], [40].

All student data used in this study are collected through an online questionnaire using Google Forms. The dataset is limited to self-reported academic history, including GPA per semester from first year up to third year as applicable; program or enrollment-related information; study-habit and engagement indicators, including study hours, LMS login frequency, and library visits; and selected financial indicators, including scholarship status or amount and family income bracket.

Family income bracket was included as a self-reported financial indicator to provide socioeconomic context in analyzing student academic standing, consistent with national reporting on household income patterns in the Philippines [41].

The dataset consists of responses from third-year students enrolled in the BSIT, BSCS, and BLIS programs under the College of Computing and Information Sciences (CCIS) of Saint Michael College of Caraga. The collected features are used to classify students as either Good Standing or At-Risk. In this study, Good Standing refers to students whose academic indicators meet the expected academic performance level, while At-Risk refers to students whose academic and engagement-related indicators suggest possible academic difficulty and the need for academic support. The predictive output is intended to support academic monitoring and early intervention planning, not to replace the professional judgment of faculty members or administrators.

Since the data are self-reported, the study acknowledges the possibility of inaccurate reporting or recall bias. Therefore, data cleaning, validation, encoding, and preparation procedures will be applied before model training to improve the reliability of the dataset.

1.2 Literature Review

This chapter presents a review of current research and technological developments related to predictive analytics in education, specifically focusing on the use of machine learning models to forecast student academic standing.

The discussion explores how Educational Data Mining (EDM) and Learning Analytics (LA) support the early identification of at-risk students and enable data-driven decision-making in higher education. It also examines the effectiveness of machine learning approaches, classification algorithms, feature preparation, model evaluation metrics, and explainable artificial intelligence in academic prediction.

The review is organized thematically into seven major sections: (1) Educational Data Mining and Learning Analytics, (2) Machine Learning in Student Performance Prediction, (3) Application of Machine Learning Classification Algorithm Models, (4) Feature Selection and Data Preprocessing, (5) Model Evaluation Metrics, (6) Explainable Artificial Intelligence, and (7) Predictive Analytics in the Local Context. The final section establishes the foundation for developing a machine learning-based predictive model tailored to the needs of Saint Michael College of Caraga (SMCC).

Educational Data Mining and Learning Analytics

In recent years, higher education institutions have increasingly adopted Educational Data Mining (EDM) and Learning Analytics (LA) to improve the monitoring of student performance and engagement. Traditional assessment methods, such as grades and instructor observations, may provide limited insights into the learning behaviors that influence academic outcomes. EDM and LA address this limitation by using educational data from learning platforms, academic records, and student activities to support data-driven decision-making and early academic intervention [1], [5], [22], [24], [37].

Aldowah et al. [5] emphasized the role of EDM and LA in supporting learning outcomes and institutional decision-making in higher education. Marcu and Danubianu [22] discussed the relationship between learning analytics and educational data mining as approaches for analyzing educational data. Palanci et al. [24] reviewed the use of learning analytics in distance education, while Zheng and Li [37] highlighted how learning analytics-informed teaching strategies can improve interactive learning in STEM education. Angeioplastis et al. [6] further demonstrated the use of educational data mining techniques for predicting student performance and enhancing learning outcomes. Recent studies also show that learning analytics can extend beyond traditional grade monitoring by incorporating behavioral indicators, educational psychology, personalized learning systems, and technology-supported instruction. Siddiqui et al. [23] examined the use of behavioral indicators in predicting student dropout risk, while Selvi et al. [32] presented an AI-based personalized learning system supported by secure analytics. Santoso et al. [29] also emphasized the role of technology in supporting instructional judgment within differentiated learning environments. These studies support the use of data-driven and technology-assisted approaches in understanding student learning conditions and academic support needs.

Machine Learning in Student Performance Prediction

Machine learning models are central to predictive analytics in education because they can identify patterns from academic, behavioral, and demographic datasets. Algorithms such as Random Forest, Support Vector Machine, Decision Tree, K-Nearest Neighbors, and Naive Bayes are commonly used to classify student performance and identify students who may be at risk of poor academic outcomes [4], [7], [8], [15], [31], [35].

Al-Ameri et al. [4] demonstrated the use of ensemble models for predicting student academic success using learning management multimedia data. Assegie et al. [7] evaluated Random Forest and Support Vector Machine models in educational data mining, while Alamri et al. [15] examined the use of Support Vector Machine and Random Forest for predicting student academic performance. Borna et al. [8] analyzed click data using AI to support student performance prediction and learning assessment. Sawarkar et al. [31] also proposed a hybrid predictive analytics model for identifying at-risk students in gamified education, and Vijayalakshmi and Srinath [35] explored the use of big data and machine learning in predicting student outcomes through cognitive behavior analysis.

Several recent studies further support the use of machine learning and predictive analytics in forecasting student academic performance across different educational contexts. Abdul Bujang et al. [2] applied machine learning and visual analytics for student grade prediction, while Edmond et al. [11] used machine learning models to analyze undergraduate student performance in computing-related courses. Green et al. [14] also demonstrated how predictive analytics can enhance learning outcomes, and Wang [36] discussed the role of machine learning in personalized learning, automated feedback, and predictive analytics in college education. Review and comparative studies by Ersozlu et al. [12], Chhimwal [13], and Tang and Sian [34] further emphasized that the performance of prediction models depends on the dataset, algorithm, and educational context. Raj et al. [26] and Ramaraj et al. [27] also explored enhanced educational data mining and deep learning techniques for forecasting student performance and behavior.

Application of Machine Learning Classification Algorithm Models

Classification algorithms play a fundamental role in student academic standing prediction because they assign students into defined categories based on patterns found in training data. In this study, the classification task focuses on identifying whether a student belongs to Good Standing or At-Risk status. The selected algorithms include Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. Previous studies show that classification performance depends on the algorithm used, dataset characteristics, preprocessing methods, and evaluation metrics. Assegie et al. [7] compared Random Forest and Support Vector Machine models in educational data mining. Alamri et al. [15] also applied Support Vector Machine and Random Forest for student academic performance prediction. Mahawar and Rattan [21] emphasized the value of ensemble machine learning approaches for early student academic performance prediction, while Kalmegh and Padar [38] discussed the importance of using appropriate evaluation metrics when comparing classification algorithms. These studies support the use of multiple classifiers in this research to determine which model performs best on the collected student dataset.

Feature Selection and Data Preprocessing

Data preprocessing and feature preparation are important steps in machine learning because the quality of input data directly affects model performance. In student performance prediction, preprocessing may include cleaning incomplete records, encoding categorical variables, scaling numerical features, and addressing class imbalance when one class has fewer samples than the other. These processes help ensure that the model can learn meaningful patterns from the dataset. Angeioplastis et al. [6] highlighted the importance of data-driven methods in predicting student performance and enhancing learning outcomes. Inyang and Johnson [16] compared XGBoost and Random Forest for predicting student academic performance, showing the relevance of model preparation and data quality in predictive tasks. Jin [17] examined student, parental, and school-related factors in academic success prediction, while Kord et al. [18] explored multimodal educational data mining techniques for course planning and performance prediction. Madahana and Ekoru [20] applied machine learning and deep learning approaches to identify at-risk students in a blended learning environment. These studies support the need for systematic data cleaning, encoding, and feature preparation before model training.

The increasing use of varied educational data highlights the need for proper feature preparation, data organization, and data consistency. Sudhan Reddy [33] discussed multimodal learning analytics, which involves the use of different data types to predict behavioral change. Kuthuru [19], although focused on enterprise AI and database systems, emphasized the importance of structured data agreements and reliable data integration, which is relevant to maintaining consistency when preparing datasets for AI-based systems. These studies support the need for systematic data cleaning, encoding, and feature engineering when academic, behavioral, and contextual indicators are combined in a predictive model.

Model Evaluation Metrics

Evaluating predictive models requires standardized metrics to determine how well each algorithm performs. In educational prediction tasks, common metrics include accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix. These metrics help measure both the general correctness of the model and its ability to identify students who may require academic intervention [4], [20], [21], [31], [38].

Al-Ameri et al. [4] used ensemble modeling to evaluate student academic success prediction. Mahawar and Rattan [21] emphasized early prediction of student academic performance using ensemble machine learning. Sawarkar et al. [31] focused on identifying at-risk students, while Madahana and Ekoru [20] applied machine learning and deep learning methods for at-risk student prediction. Kalmegh and Padar [38] discussed the role of evaluation metrics in comparing classification algorithms. These studies support the use of accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix in evaluating the models used in this study.

Explainable Artificial Intelligence

Despite their predictive capability, machine learning models are often criticized as "black boxes" because their internal decision-making processes are not always easily understood by users. Explainable Artificial Intelligence (XAI) helps address this limitation by providing interpretable explanations of model predictions. SHAP explains predictions by assigning feature-importance values to model outputs, while LIME explains individual predictions by approximating the model locally with an interpretable explanation model [39], [40]. In educational settings, interpretability is important because teachers and administrators need to understand why a student is classified as Good Standing or At-Risk before making academic intervention decisions.

Aldowah et al. [5] emphasized the importance of transparency and data-driven decision-making in educational data mining and learning analytics. Borna et al. [8] highlighted the use of AI-based analysis of student learning data to improve student performance prediction and learning assessment. Chauhan and Mishra [10] presented an intelligent educational system that applies learning analytics and visualization to support student performance prediction. Pektaş et al. [25] also discussed the role of educational data mining and AI-supported learning analytics in improving instructional decision-making. These studies support the integration of explainable results in predictive systems because interpretable outputs can help educators trust and responsibly use prediction results.

Predictive Analytics in the Local Context

Localized implementation of predictive analytics is important for institutions such as Saint Michael College of Caraga (SMCC) because student characteristics, available data, academic policies, and institutional resources may differ across educational settings. Predictive models developed using local student data are more appropriate for institutional decision-making because they reflect the actual academic environment of the target school.

Caday et al. [9] demonstrated the use of predictive analytics in analyzing student academic performance, showing that educational data can support academic monitoring and decision-making. Saragih [30] applied Random Forest to predict student academic performance using LMS behavioral data. These studies support the development of a localized student academic standing prediction system that uses institutional data, machine learning models, and explainable results to assist SMCC in identifying students who may need academic support.

1.3 Project Objectives

General Objective

To develop a predictive system that uses multiple machine learning models to classify student academic standing, providing interpretable results and actionable insights for educational decision-making.

Specific Objectives

1. To analyze requirements and identify the necessary data sources, system functions, and user needs for the predictive model.

2. To design a functional system that integrates survey-collected student data for academic standing classification as either Good Standing or At-Risk.

3. To train and test multiple machine learning models, including Decision Tree, Random Forest, SVM, KNN, and Naive Bayes, incorporating Explainable Artificial Intelligence (XAI) techniques for model interpretability and transparency.

4. To evaluate the performance of each model using accuracy, precision, recall, F1-score, and ROC-AUC metrics.

5. To assess the system's functionality, performance efficiency, usability, and reliability based on ISO 25010 standards to ensure software quality.

1.4 Scope and Limitations

This study focuses on the development of a predictive analytics system that employs machine learning models, to forecast student academic standing at Saint Michael College of Caraga, particularly within the College of Computing and Information Sciences (CCIS). The dataset consists of responses from third-year students enrolled in BSIT, BSCS, and BLIS programs who voluntarily completed a Google Forms questionnaire. The model classified students as Good Standing or At-Risk based on self-reported academic history (GPA per semester from first year up to third year, as applicable), study-habit and engagement indicators (study hours, LMS login frequency, and library visits), and self- reported financial indicators (scholarship status/amount and family income bracket). The questionnaire also collected self-reported learning behavior and motivation indicators, including academic satisfaction, time management, study regularity, course motivation, submission punctuality, and subject interest. The predictive results were presented through a decision-support dashboard for administrators and faculty to aid in early academic intervention and retention planning. No official institutional databases (e.g., Registrar or LMS back-end records) are directly accessed; the study relies on participant-provided responses.

The model is limited to datasets collected from SMCC, which may affect generalizability to other institutions. Incomplete, inconsistent, or biased data may influence prediction accuracy. The system does not include non-academic factors such as mental health, employment status, or internet access, which also affect performance. Ethical considerations—such as data privacy, bias, and potential misclassification—are recognized, and measures were taken to ensure confidentiality and responsible use. The prototype does not yet support long-term tracking or qualitative validation, and results are intended to assist, not replace, academic judgment.

1.5. Definition of Terms

In this study, the following terms are defined operationally based on their application in developing and evaluating machine learning models for student academic performance at Saint Michael College of Caraga:

Academic Standing - Refers to the classification of students at Saint Michael College of Caraga based on their overall academic performance. In this study, students are categorized as Good Standing, or At-Risk according to the predictive output of the machine learning models.

Accuracy - The share of predictions the machine learning models gets right out of all predictions made, giving a general sense of how well the classifier identifies students' academic standing overall. While useful as a headline metric, it is best interpreted alongside precision, recall, and the confusion matrix for a more comprehensive performance evaluation.

At-Risk - A classification output from the predictive model indicating students with a high likelihood of academic failure or dropout. These students are prioritized for academic counseling and targeted intervention based on their predicted risk level.

Confusion Matrix - A tabular representation used to evaluate the model's performance by comparing predicted outcomes with actual results. It shows how many instances were correctly and incorrectly classified for each category of academic standing.

Data Cleaning - A crucial step in data preprocessing that involves detecting, correcting, or removing inaccurate, incomplete, or inconsistent data entries from the dataset to ensure high-quality and reliable model training results.

Data Preprocessing - The process of cleaning, transforming, and organizing raw survey data (Google Forms responses) to prepare it for model training. This includes handling missing values, encoding categorical variables, and balancing class distributions when needed to improve prediction accuracy.

Decision-Support System - The functional component of the study that presents predictive results through an interactive dashboard. It enables faculty and administrators to visualize model outcomes, monitor student performance, and make timely, data-driven academic decisions.

Educational Data Mining (EDM) - The process of applying data mining techniques to educational datasets to uncover hidden patterns and trends. In this study, EDM is used to identify academic, study-habit/engagement, and socioeconomic factors from survey data that affect student performance and academic standing.

Explainable Artificial Intelligence (XAI) - A framework of methods and tools that make the predictions of machine learning models understandable to humans. The study integrates XAI to

help educators interpret the model's reasoning behind each academic standing classification, ensuring transparency and trust.

F1-Score - A model evaluation metric that combines precision and recall into a single measure, representing their harmonic mean. It helps determine the balance between correctly identifying at-risk students and minimizing false classifications.

Feature Selection - The process of determining which input variables contribute most significantly to predicting student academic performance. It enhances model interpretability and reduces computational complexity by eliminating irrelevant or redundant features.

Good Standing - A predictive label assigned to students who meet or exceed institutional academic standards. Students in this category exhibit consistent performance and require minimal intervention from academic advisers.

Learning Analytics (LA) - The measurement and analysis of student learning behaviors using data-driven techniques. Within this project, LA supports the early identification of performance trends and assists in developing strategies for improving academic outcomes.

Machine Learning (ML) - A subset of artificial intelligence that enables computers to learn patterns from data without being explicitly programmed. This study applies ML techniques, to predict students' academic standing based on the collected survey dataset.

Precision - A metric that measures the proportion of correctly predicted instances among all instances predicted for a specific category. In this study, it shows how accurately the model identifies students as Good Standing, or At-Risk without false positives.

Predictive Analytics - The overarching framework of this study, involving statistical and machine learning methods to forecast student academic standing. It enables the institution to identify potential academic risks and implement early interventions.

Recall - A model evaluation metric that measures the proportion of actual instances of a class (for example, all At-Risk students) that the model correctly identifies. It reflects the model's sensitivity in detecting students who truly require academic intervention.

ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) - A performance metric that evaluates the model's ability to distinguish between different classes. A higher ROC-AUC value indicates a stronger predictive capability in classifying students accurately across all categories.

SMOTE (Synthetic Minority Oversampling Technique) - A data-balancing method used during preprocessing to address class imbalance by generating synthetic examples of underrepresented categories, such as the At-Risk class, to improve model fairness and accuracy.

Student Retention - The institutional objective of keeping students enrolled and progressing toward graduation. The predictive model contributes to this goal by identifying students at risk of academic failure early, allowing for proactive support measures.

CHAPTER 2

2.1 Research Design

This study employs a quantitative, developmental research design. The quantitative approach is essential for the statistical analysis of survey-collected student data, which include academic history (self-reported GPA per semester), study-habit and engagement indicators (study hours, LMS login frequency, and library visits), and selected financial indicators (scholarship status/amount and family income bracket). This design allows for the application of machine learning techniques to identify patterns and build a predictive model for academic standing. The developmental aspect of the research involves the systematic design, creation, and evaluation of the predictive system as a software artifact.

2.2 Software Development Life Cycle

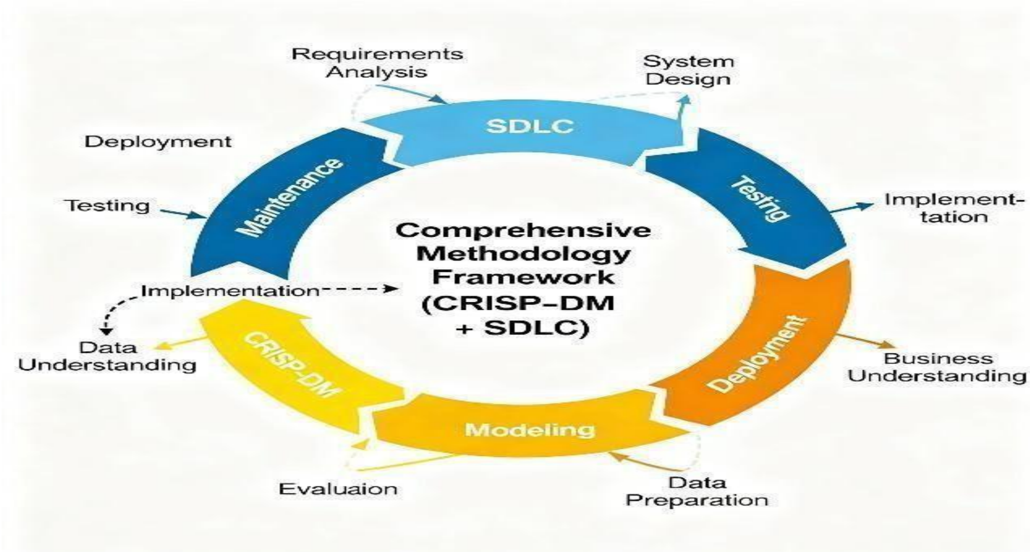


Figure 1: Comprehensive Methodology

This study employs a comprehensive methodology that synergizes the structured phases of an iterative Software Development Life Cycle (SDLC) with the cyclical, data-centric phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM). This integrated framework provides a robust structure for managing both the software engineering and data science components of the project. The SDLC governs the overall project architecture, from requirements analysis to final system deployment, while the CRISP-DM framework is nested within it to guide the iterative processes of data understanding, preparation, modeling, and evaluation. This dual approach ensures that the development of the predictive model is both methodologically sound and aligned with the software implementation goals.

2.3 System Architecture

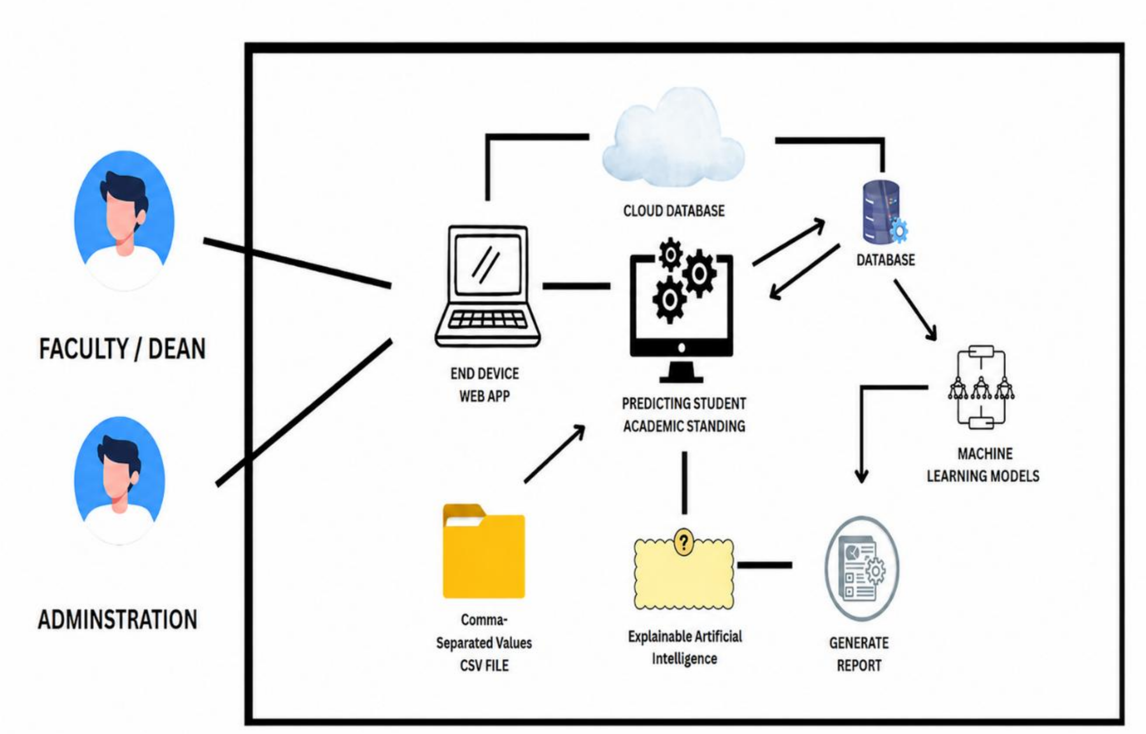


Figure 2: System Architecture

Figure 2 illustrates the system is designed using a multi-tier web architecture, a standard paradigm for developing robust and scalable web-based applications. This architecture logically separates the system into three distinct layers: the Client-Side, the Server-Side, and the Data Tier. This separation ensures modularity, simplifies maintenance, and enhances security by isolating the business logic and data storage from the user interface.

2.4 Conceptual Framework

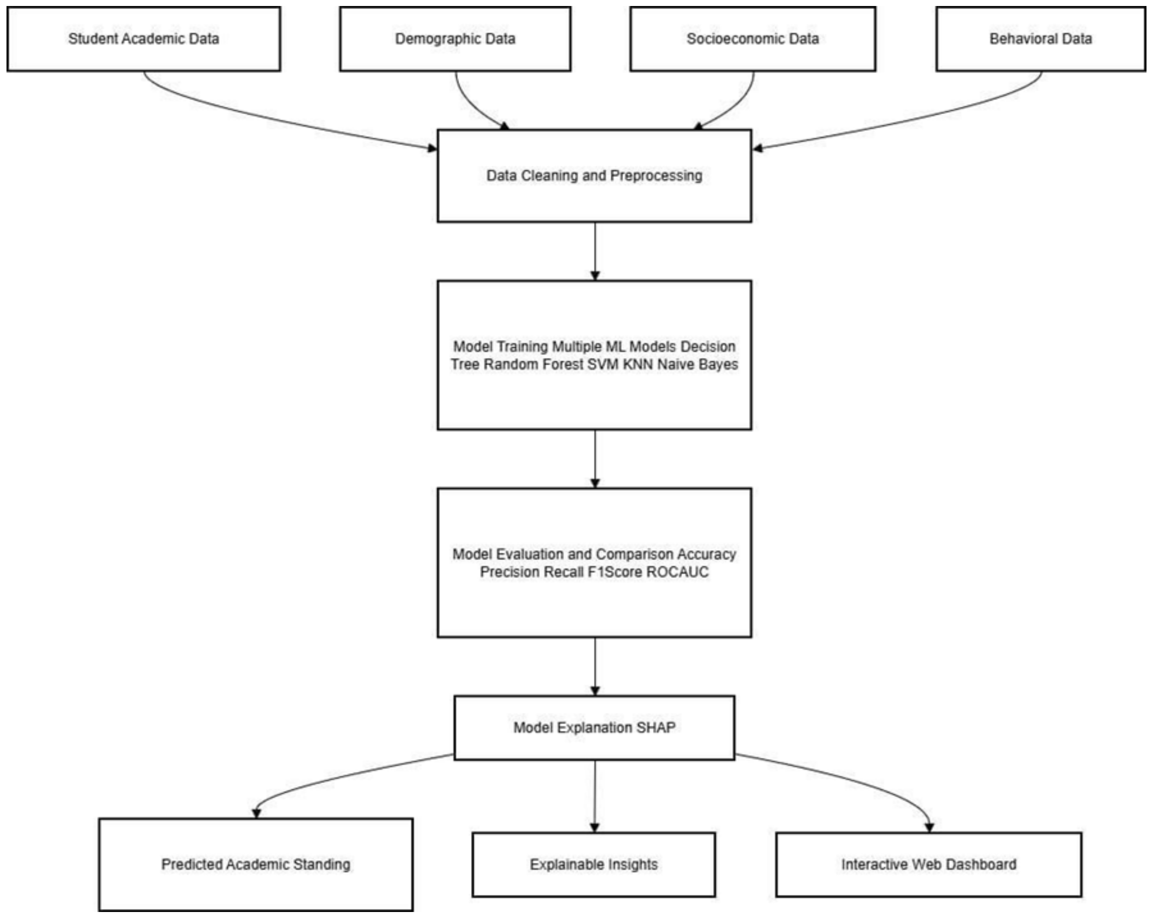


Figure 3: Conceptual Framework

Figure 3 conceptual framework of this study is illustrated using an Input-Process- Output (IPO) model. This model provides a clear, high-level overview of the research flow, demonstrating how raw data is systematically transformed into actionable intelligence for educational stakeholders. The framework is composed of three sequential stages: the Input, the Process, and the Output. In the process stage, the cleaned data are used to train and compare multiple machine learning algorithms, and the best-performing model is selected to generate academic standing predictions and explainable insights for stakeholders.

2.5 Algorithms / Framework Used

This study utilizes multiple machine learning algorithms to predict students' academic standing. These algorithms enable comparative analysis and selection of the most effective predictive model for educational data. The framework integrates (1) data-driven academic factors, (2) model training and prediction, and (3) measurability through standard classification performance metrics to ensure reliability of results.

Machine Learning Framework and System Workflow

The system follows a supervised learning classification framework where measured academic factors are transformed into features used to predict the student's academic standing category.

System Workflow:

Data Collection (Academic Factors Input)

Student predictors are collected through the Google Forms questionnaire and include self-reported academic history (GPA per semester from first year up to third year, as applicable), study-habit and engagement indicators (study hours per day, LMS login frequency per week, and library visits per month), and selected financial indicators (scholarship status/amount and family income bracket). Each response is converted into measurable values (e.g., numeric GPA values and coded categorical ranges) and prepared for preprocessing, storage in the system database, and model training

Preprocessing and Feature Preparation

Data are cleaned to handle missing or inconsistent records. Categorical variables (if present) are encoded, and numerical variables may be scaled/normalized as needed, especially for algorithms such as SVM and KNN. The output of this stage is a feature vector x representing each student.

Model Training and Validation

The dataset is split into training and testing sets (or validated through cross-validation). The following machine learning algorithms are trained using the same feature set: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes.

Prediction and Classification Output

For every student record, the trained model generates a predicted academic standing class $\hat{C}$. The system outputs the prediction for decision support in identifying students who may require intervention.

Performance Evaluation (Measurability)

The system produces a confusion matrix and computes standard evaluation metrics to measure how effectively each model predicts each academic standing category. These metrics guide the selection of the best model for deployment.

Algorithms Used

Decision Tree Algorithm

The Decision Tree algorithm is employed due to its interpretability and ability to model non-linear relationships. It operates by recursively partitioning the feature space into decision nodes based on attribute values that minimize an impurity measure. For classification tasks, impurity is commonly measured using Gini impurity, calculated as:

$G = 1 - \Sigma (p_k^2)$, for k = 1 to K

where p_k represents the proportion of class k in a node and K denotes the total number of classes. Decision Trees provide transparent decision rules that aid in understanding how student attributes influence academic outcomes.

Role in the system: Generates human-readable IF–THEN rules that support explainability in academic decision-making.

Random Forest Algorithm

The Random Forest algorithm can handle complex and multidimensional educational datasets. Random Forest operates by constructing an ensemble of decision trees ($T_1, T_2, \ldots, T_k$), each trained on bootstrapped samples of the training data with random feature selection at each split to increase diversity.

For regression tasks, the predicted outcome $\hat{y}$ for an input vector x is computed as the mean prediction across all trees:

$\hat{y} = (1/M)\ \Sigma\ T_j(x)$, for j = 1 to M

For classification tasks, the predicted class $\hat{C}$ is obtained through majority voting of individual tree predictions:

$\hat{C}$ = mode $\{T_1(x), T_2(x), \ldots, T_k(x)\}$ This ensemble learning strategy improves generalization performance and predictive stability, making Random Forest well suited for educational prediction tasks.

Role in the system: Produces stable predictions using multiple trees and supports ranking of influential academic factors.

Support Vector Machine (SVM) Algorithm

The Support Vector Machine algorithm is applied due to its effectiveness in high-dimensional classification problems. SVM identifies an optimal decision boundary that maximally separates data points of different academic standing categories. The algorithm focuses on support vectors, which are the data points closest to the boundary, enhancing robustness and generalization in complex student datasets.

Role in the system: Separates academic standing categories using a margin-maximizing boundary; feature scaling is typically applied to ensure fair contribution of all academic factors.

K-Nearest Neighbors (KNN) Algorithm

The K-Nearest Neighbors algorithm is included due to its simplicity and effectiveness in pattern recognition. KNN classifies a student by identifying the k closest instances in the dataset using a distance-based measure. The predicted academic standing is determined by majority voting among the nearest neighbors. This approach enables the model to capture similarities among students with comparable academic and behavioral characteristics.

Role in the system: Predicts academic standing by comparing a student's factor profile to similar historical cases; scaling/normalization is often applied for fair distance computation.

Naïve Bayes Algorithm

The Naive Bayes algorithm is utilized due to its computational efficiency and probabilistic foundation. It estimates the probability of a student belonging to a particular academic category based on observed features and prior probabilities. Although it assumes feature independence, Naive Bayes performs effectively in many educational classification tasks and serves as a reliable baseline model.

Role in the system: Provides probability-based predictions and functions as a benchmark model for comparison.

Explainable Artificial Intelligence (XAI) Framework

Explainable Artificial Intelligence (XAI) is a framework designed to make machine learning model more transparent and interpretable. It encompasses a variety of algorithms, such as SHAP and LIME, that quantify and visualize how individual features influence model predictions. In this study, SHAP is integrated as the primary XAI method for explaining machine learning model outputs.

To enhance interpretability and stakeholder trust, the SHAP (SHapley Additive exPlanations) framework is incorporated to decompose individual model predictions into additive feature contributions ϕ_i. SHAP values are grounded in cooperative game theory, providing a mathematically principled method for attributing the effect of each feature.

The prediction model f(x) for an instance x can be expressed as:

$$f(x) = E[f(X)] + \Sigma_{i=1}^{M} \phi_i$$

where E[f(X)] is the expected output of the model over the dataset and M is the total number of input features. The SHAP values ϕ_i quantify each feature's marginal contribution averaged over

all possible feature combinations. SHAP enables clear, consistent explanations of complex model outputs, supporting actionable insights for targeted academic interventions.

Model Evaluation Metrics

The model's predictive performance is assessed using standard binary classification metrics. In this study, the system classifies students into two academic standing categories: Good Standing and At-Risk.

Accuracy measures the proportion of correct predictions:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision assesses the correctness of positive predictions:

Precision = TP / (TP + FP)

Recall measures the proportion of actual At-Risk students correctly identified by the model:

Recall = TP / (TP + FN)

F1-score provides the harmonic mean of precision and recall, balancing false positives and false negatives:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Here, TP, TN, FP, and FN signify true positives, true negatives, false positives, and false negatives, respectively. The Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) is also used to evaluate the model's ability to distinguish between Good Standing and At-Risk students across classification thresholds.

In this study, recall and F1-score for the At-Risk class are emphasized because the system aims to minimize false negatives and support early academic intervention.

2.6 Use Case Diagram

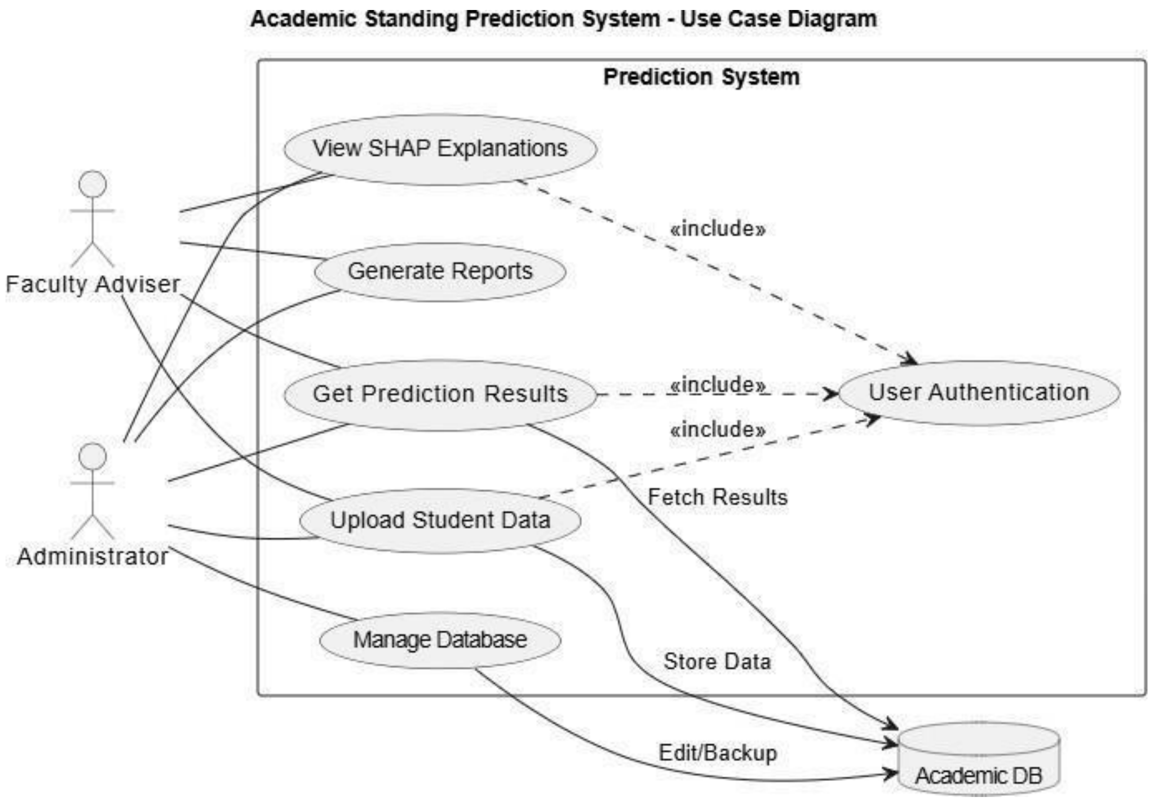


Figure 4: Use case Diagram

Figure 4 illustrates how administrators and faculty advisers interact with the system's core functions, with user authentication securing all critical actions and database interactions ensuring centralized and regulated data management.

2.7 Activity Diagram

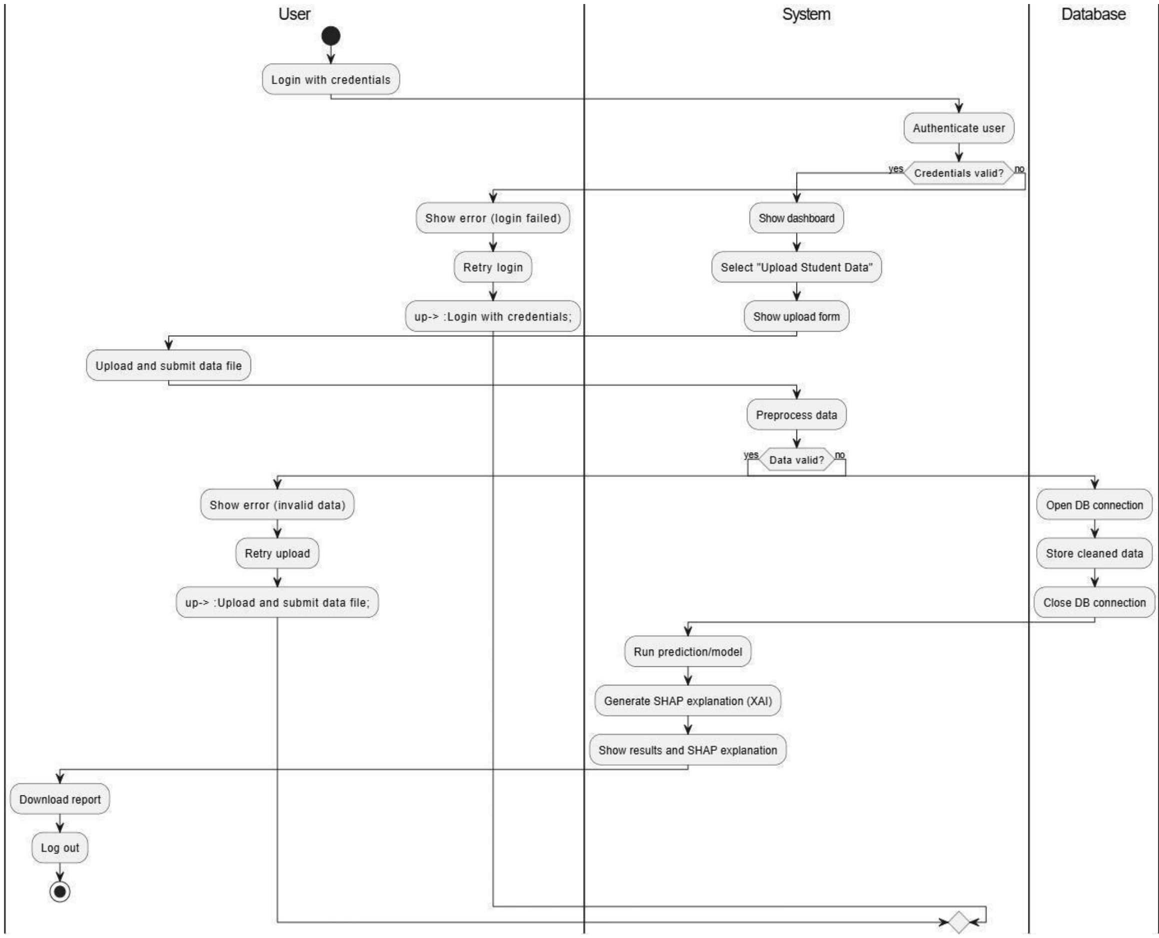


Figure 5: Activity Diagram

Figure 5 illustrates how administrators and faculty advisers interact with the prediction system's main workflow, including secure login, data upload, validation, and centralized database operations. After generating prediction results, the system produces SHAP- based explanations, ensuring users receive both outcomes and model reasoning. Decision nodes manage feedback and retries, while swimlanes delineate the responsibilities of users, system, and database throughout the process. The inclusion of SHAP explanations enhances transparency and interpretability for all end users.

2.8 Sequence Diagram

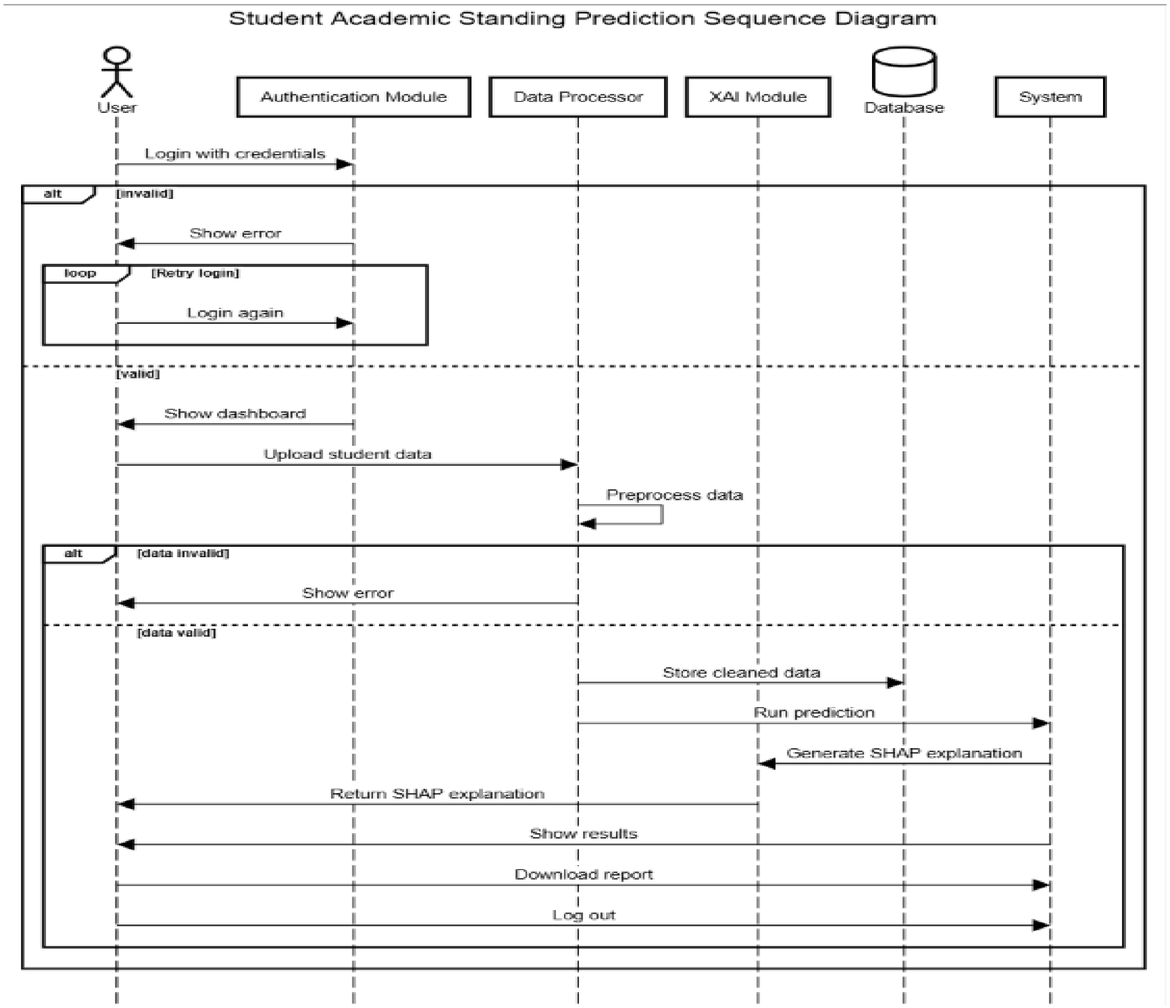


Figure 6: Sequence Diagram

Figure 6 shows how users interact with the prediction system to log in, upload and validate student data, and store it in the centralized database. The system then generates prediction results along with SHAP explanations, providing both outcomes and model reasoning. Conditional branches handle errors and retries, while distinct lifelines represent the roles of users, system modules, and the database. SHAP explanations ensure transparency and interpretability of the predictions.

2.9 ERD (Database Design)

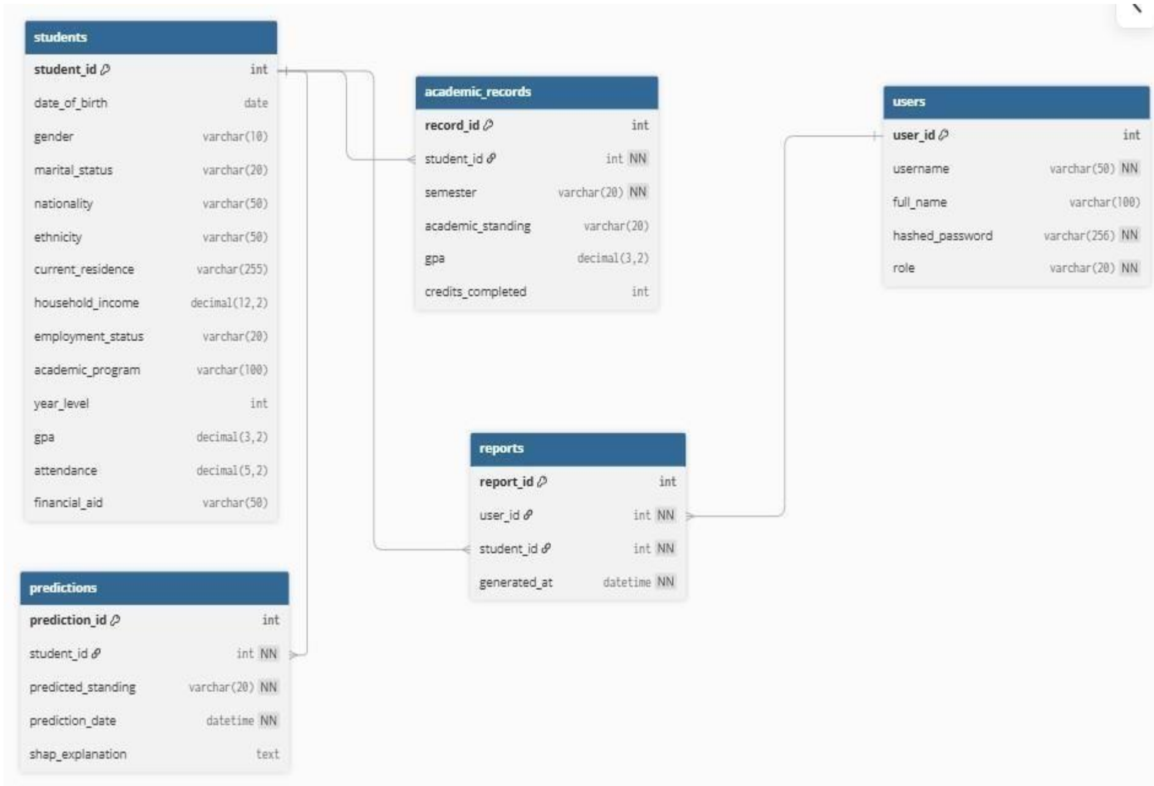


Figure 7: Entity Relationship Diagram

Figure 7 shows the core tables and relationships for managing students, academic records, predictions, user accounts, and reports. Each student is linked to their records and prediction results, while users can generate reports. This design ensures organized, secure data access for prediction and reporting in the academic system.

2.10 User Interface Design (Prototype)

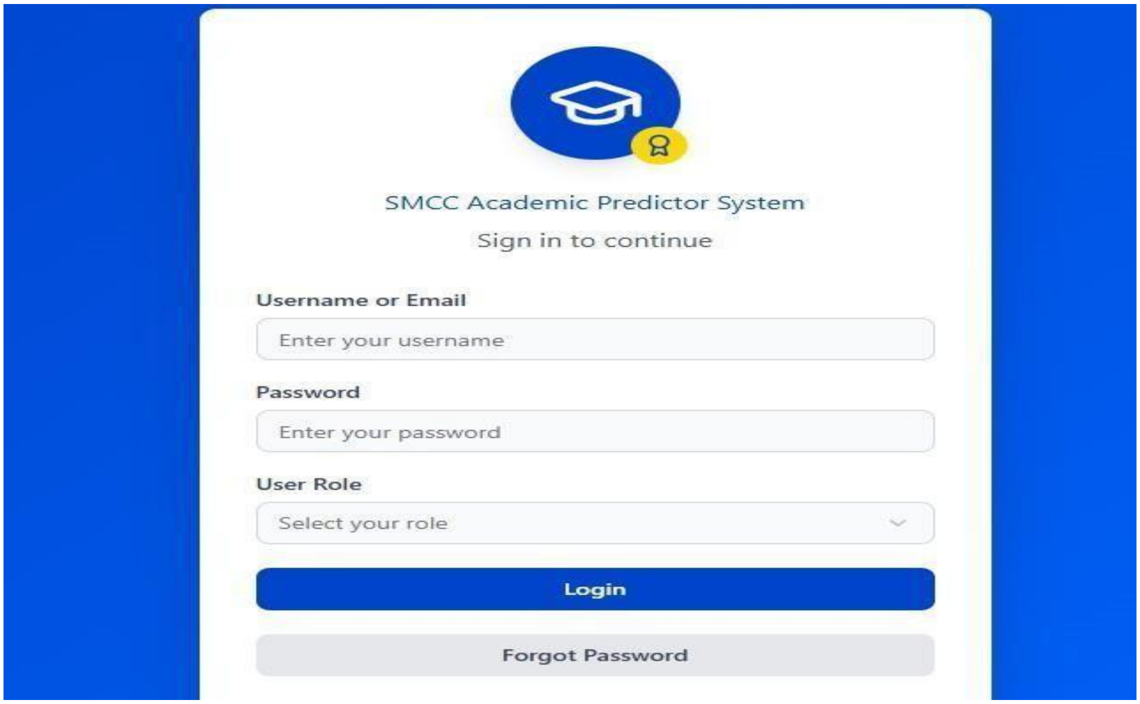


Figure 8: Login Screen

Figure 8 shows how users interact with the prediction system to log in, select their role, and authenticate access to the centralized application. The login form establishes user identity as the gateway to uploading data, generating predictions, and viewing model explanations. Conditional branches support error handling for invalid credentials, while clear lifelines distinguish authorized user roles and secure access, ensuring interpretability and integrity for all subsequent processes.

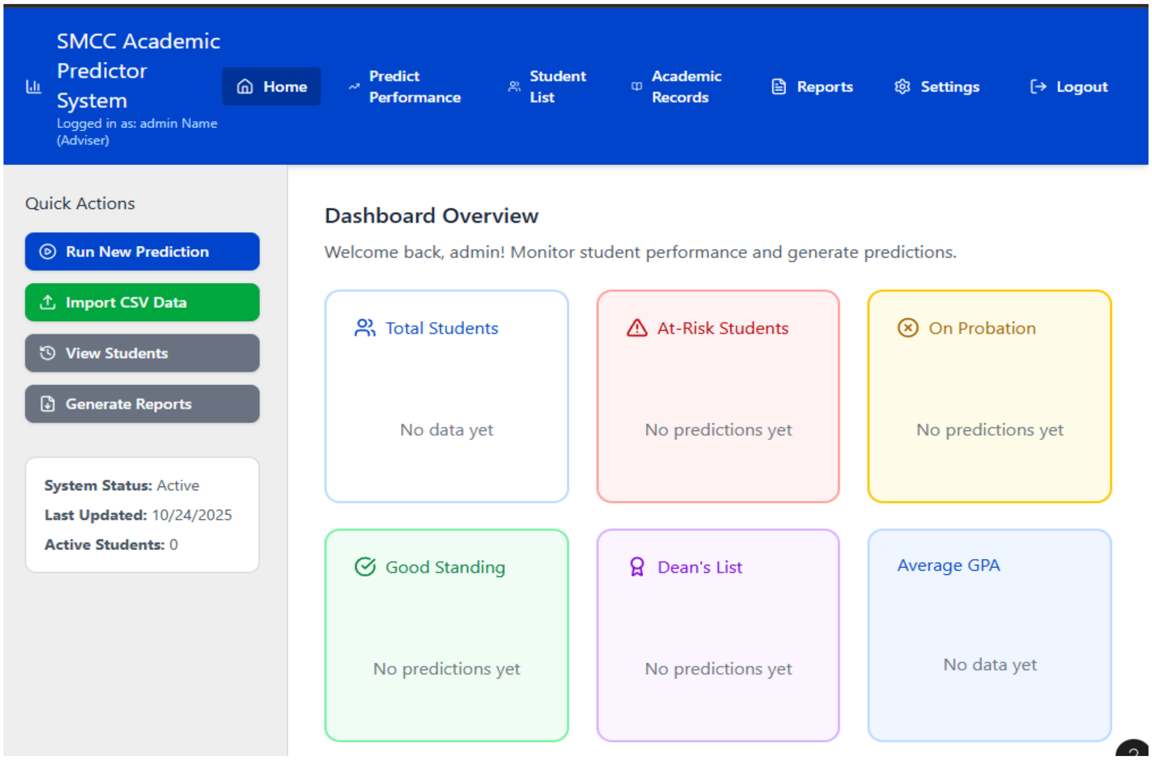


Figure 9: Dashboard/Home

This figure 9 shows how users interact with the prediction system dashboard to view summary metrics, monitor prediction outcomes, and access core features such as running new predictions or importing CSV data. Each dashboard card and quick action links users to data upload, model execution, and result validation. Conditional branches handle empty-state messaging and errors, while key indicators and SHAP-driven output provide a transparent, interpretable overview of prediction results.

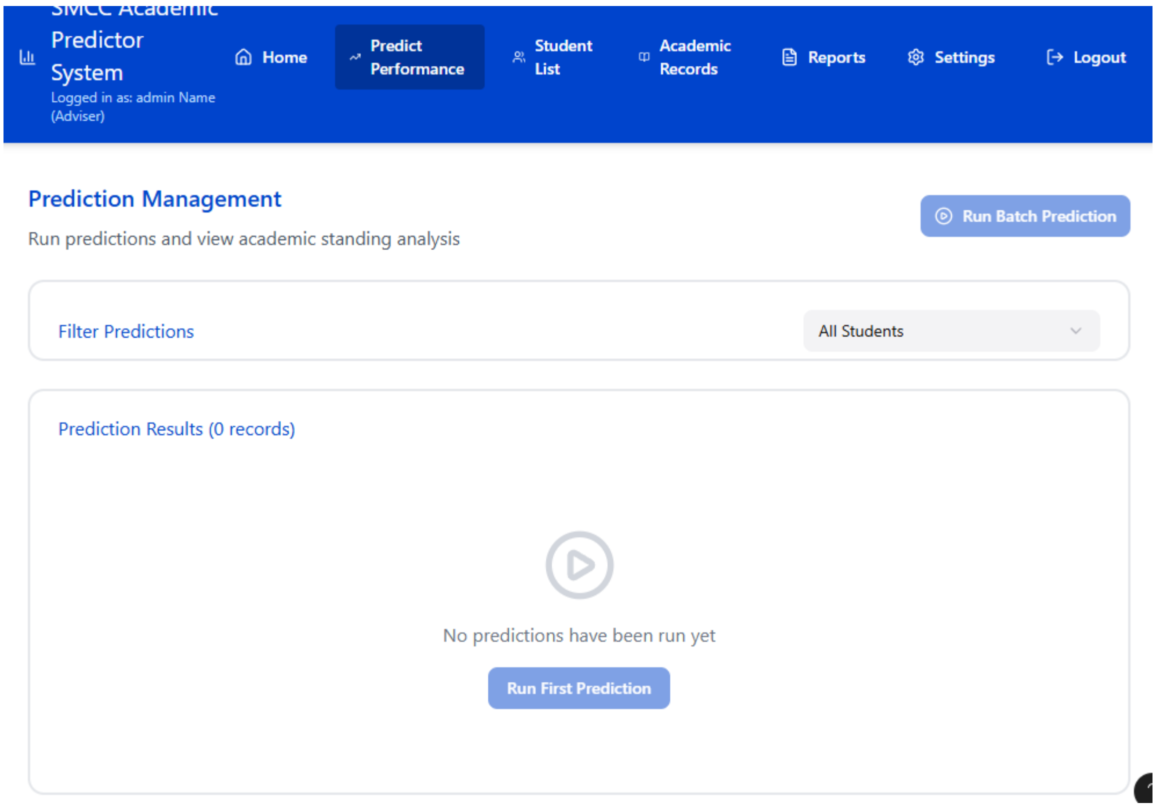


Figure 10: Prediction Management

Figure 10 shows how users interact with the prediction system to manage and execute academic standing predictions. Users can run batch predictions, filter by students, and review prediction results alongside SHAP explanations. The system ensures that both outcomes and reasoning are displayed, supporting transparency and model interpretability. Conditional branches facilitate retries on failures, while distinct interface lifelines support streamlined prediction tasks and XAI visualization.

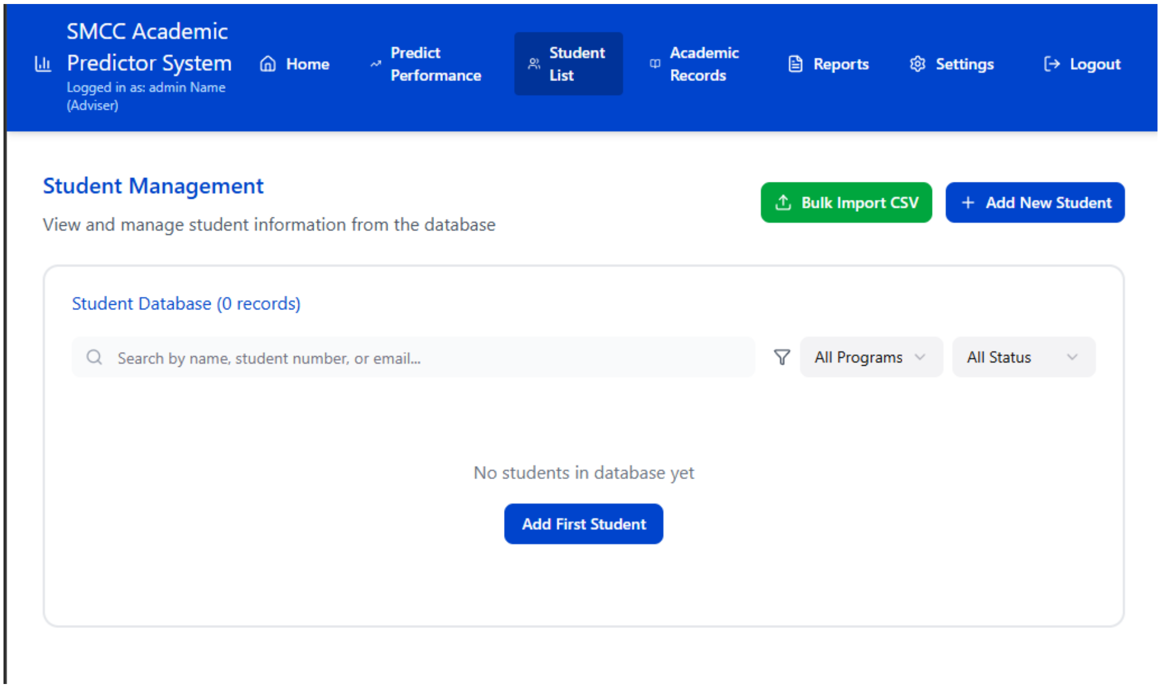


Figure 11: Student Management

This figure shows how users interact with the system to manage the student database, including adding new students and bulk importing via CSV. The management module supports search, filters, and status updates, integrating directly with the centralized database for validated storage. The workflow includes error handling and retry logic, and connects to prediction modules where SHAP explanations ensure interpretability for each student's predicted standing.

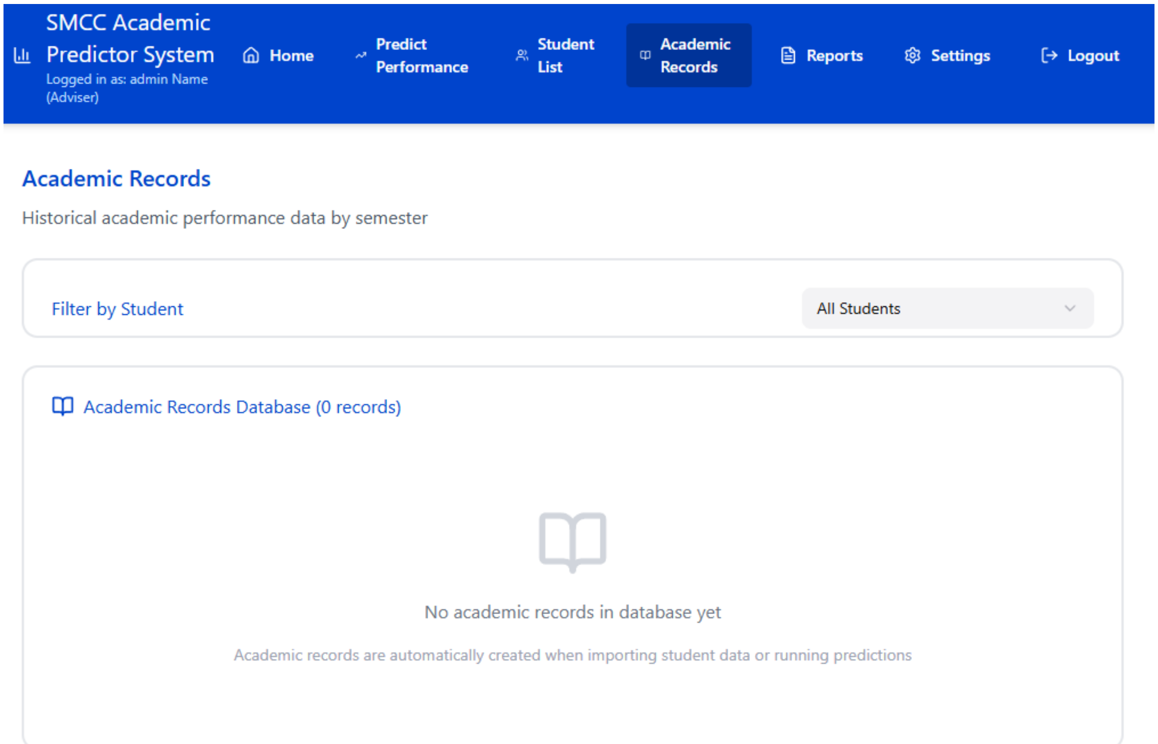


Figure 12: Academic Records

This figure shows how users interact with the prediction system to filter, review, and manage historical academic records by student and semester. Academic records are synchronized with the centralized database and automatically updated by data import and prediction activity. The interface supports conditional messaging for empty data states, while integrating with prediction and SHAP explanation workflows to maintain full transparency and interpretability of academic analysis.

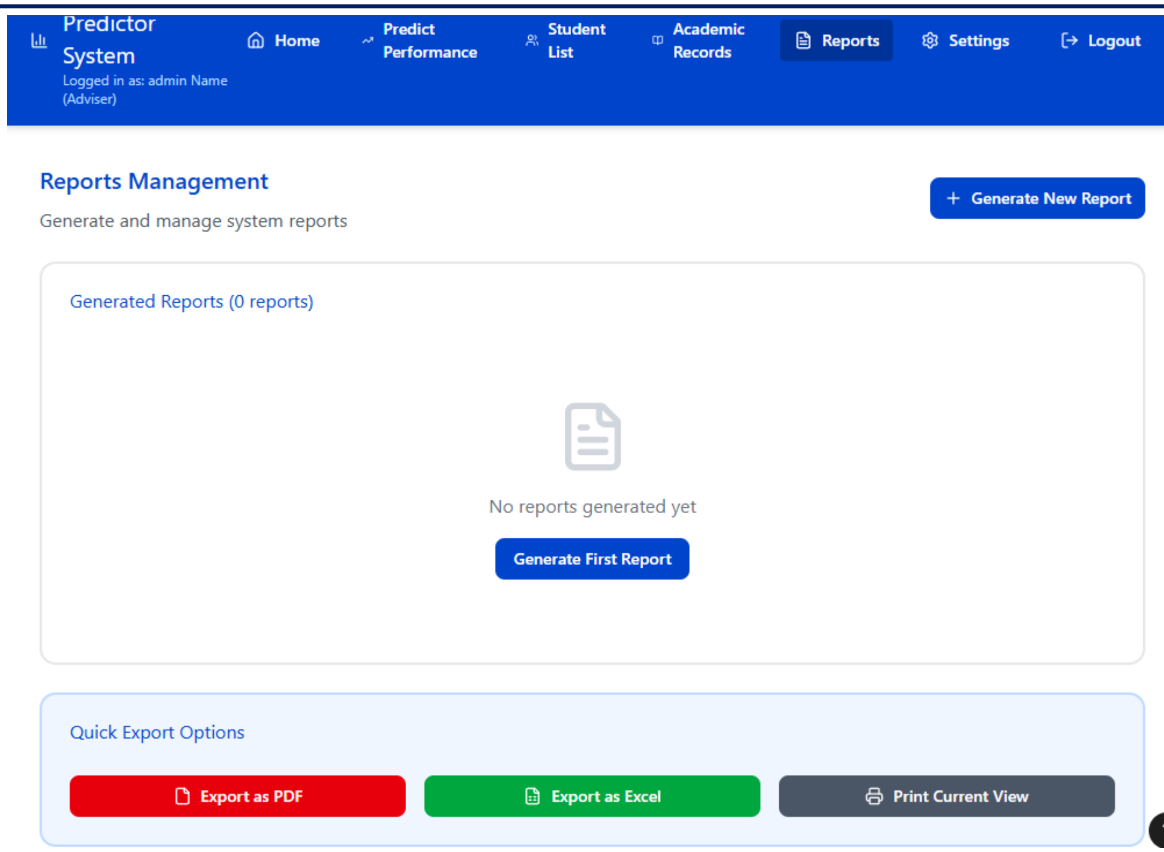


Figure 13: Reports Management

This figure shows how users interact with the prediction system to manage generated reports, export data, and print system views. The reporting interface enables users to create, review, or export results, with the system highlighting error states when no reports exist. Lifeline separation distinguishes reporting, database storage, and user operations, with SHAP explanations integrated into report insights for enhanced transparency and interpretability.

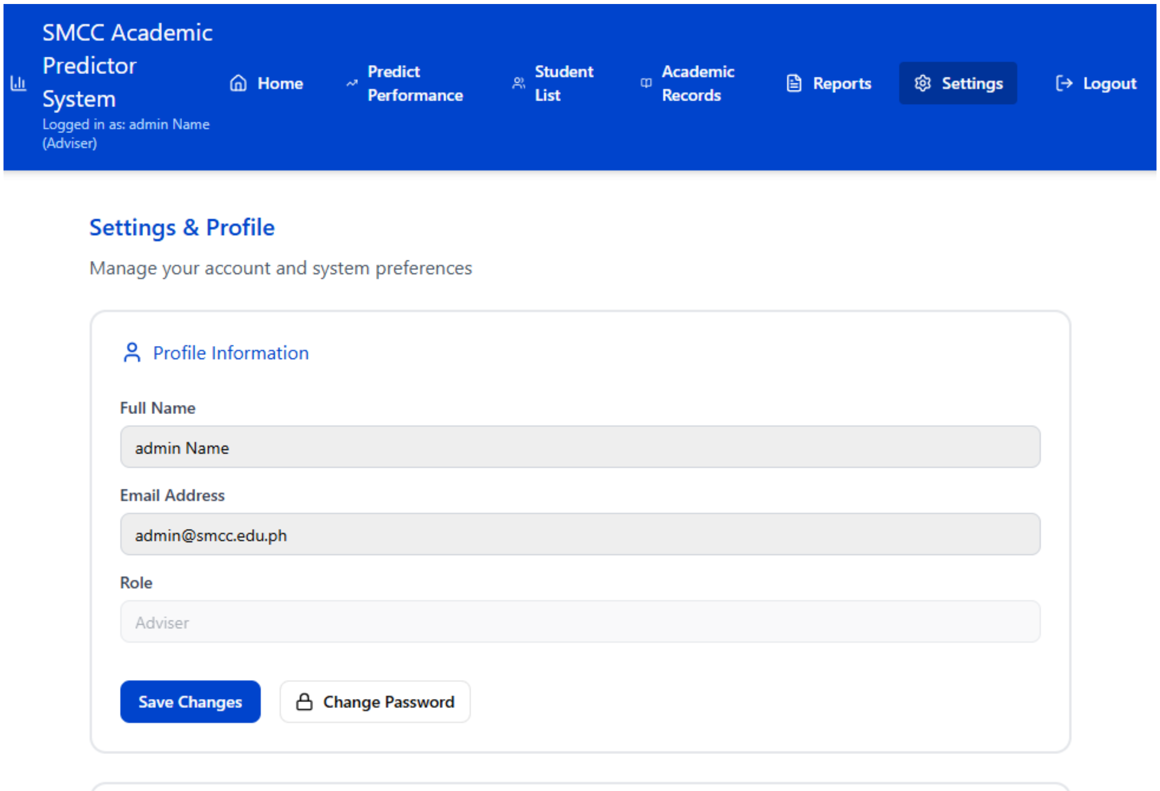


Figure 14: Settings & Profile

This figure shows how users interact with the prediction system to manage their personal account details, update credentials, and review role-based information. Profile management safeguards user authentication and system preferences, with conditional logic handling changes and validation. Distinct system lifelines ensure that user settings are maintained securely and accessed appropriately, supporting traceability and proper data stewardship throughout predictive and explanatory workflows.

2.11 Software Platforms, Development Environments and Tools

Table 1

Software Requirements

Component	Specification	Usage
Front-end		
Jupyter Notebook	JupyterLab (Anaconda)	Interactive Python coding, data visualization, and EDA.
Streamlit (optional)	Python Web UI Framework	Dashboard for presenting prediction results and user interactions.
HTML/CSS/JS (optional)	Web Standards	For custom visualization dashboards if web deployment is required.
Back-end		
Python	3.8+	Develops entire ML pipeline: preprocessing, modeling, automation scripts.
Flask/FastAPI (optional)	Python Web Framework	Provides REST API endpoints for serving predictions in a web environment.
scikit-learn	Python Library	Machine Learning algorithm, preprocessing, validation, and evaluation.

pandas	Python Library	Data cleaning, transformation, and merging.
NumPy	Python Library	Matrix and vector operations for efficient computations.

SHAP, LIME	Python Library	Model interpretation and feature analysis.
Matplotlib, Seaborn	Python Libraries	Plots data distributions and evaluation metrics.
Database/Storage		
MySQL/PostgreSQL	RDBMS	Stores survey responses (Google Forms), derived features, prediction outputs, SHAP explanations, and system user/audit records for the prototype.
Excel/CSV	File Format/Tool	Data transfers, backup, and initial cleaning.
Version Control/Cloud		
Git, GitHub	Version Control	Tracks code development and team collaboration.
Google Colab/Kaggle	Cloud Notebooks	Cloud computation, easy sharing and scalable model training.

2.12 Hardware Requirements

Table 2

Hardware Requirements

Components	Specification	Usage
Development PC		
Processor	Intel Core i5 (8th gen) or AMD Ryzen 5	For running data analysis, training models, and development tasks.
RAM	8 GB DDR4	Supports multitasking and fast computation during training and EDA.
Storage	256 GB SSD	Quick access and ample space for large datasets and library dependencies.
Display	1366x768 resolution or higher	Ensures clear visibility of data visualizations and code interfaces.
OS	Windows 10/11, Ubuntu 20.04+, or macOS	Compatibility with necessary ML libraries and tools.

Internet Access	Broadband (10 Mbps+)	Enables data collection, cloud computation, collaboration, and API access.

Peripheral Devices	Mouse, Keyboard, Webcam	Standard input and communication devices for remote collaboration/reporting.

2.13 System Evaluation/Evaluation Procedure

Software Evaluation Using ISO 25010 Standards

The system was evaluated based on selected ISO/IEC 25010 software product quality characteristics. Although ISO 25010 defines eight software quality characteristics, this study focused only on the criteria aligned with the fifth specific objective of the study, namely functional suitability, performance efficiency, usability, and reliability.

The following criteria were used in evaluating the system:

- **Functional Suitability** – Measures how well the system provides functions that meet the stated needs, including completeness, correctness, and appropriateness.
- **Performance Efficiency** – Assesses system response time and processing efficiency during system use.
- **Usability** – Evaluates how easily users can understand, learn, navigate, and operate the system.
- **Reliability** – Evaluates the system's consistency, stability, and ability to perform its intended functions during operation.

The evaluation was conducted using a researcher-made questionnaire based on the selected ISO 25010 criteria. The responses were analyzed using weighted mean and interpreted using a five-point verbal interpretation scale.

AI/ML Model Evaluation Metrics

For the predictive analytics component of the study, the performance of the machine learning models will be evaluated using standard classification metrics. These metrics will be used to determine how well the models classify students as **Good Standing** or **At-Risk**.

- Accuracy: The overall percentage of correctly classified instances among all predictions.
- Precision: The proportion of true positive predictions among all positive predictions for a given class.
- Recall : The proportion of actual positives correctly identified by the model.
- F1-Score: The harmonic mean of precision and recall, providing a balanced measure for imbalanced datasets.
- ROC-AUC (Receiver Operating Characteristic - Area Under the Curve): Evaluates the model's ability to distinguish between classes.
- Confusion Matrix: Reports the counts of true positives, false positives, true negatives, and false negatives for all categories, supporting granular performance analysis.

By combining the selected ISO 25010 criteria for software quality evaluation with AI/ML model evaluation metrics, the study will assess both the quality of the developed system and the predictive performance of the machine learning models.

2.14 Research Ethical Standards

The development and implementation of the Student Academic Standing Prediction System strictly follow established ethical standards to ensure data security, participant protection, and research integrity. The system is designed with respect for the rights, privacy, and wellbeing of all involved parties, maintaining transparency and academic rigor throughout the study.

A. Protection of Intellectual Property Rights (IPR)

This study recognized intellectual property rights by protecting the researchers original outputs, including the survey instrument, documentation, source code, system prototype, database design, and machine learning workflow. These materials were intended solely for academic and non-commercial research purposes and were protected under applicable copyright principles. The study used third-party software libraries, frameworks, and reference materials under their respective licenses, and all sources were properly cited in compliance with open-source and copyright requirements. The developed system and machine learning framework were not patented at this stage, and no commercial product or proprietary technology is was claimed from this research.

B. Informed Consent

Participation in this study was voluntary, and digital informed consent was obtained through the Google Forms questionnaire before any participant could access the survey. Participants were informed of the study purpose, procedures, expected duration, types of data to be collected, possible minimal risks and benefits, confidentiality measures, and their rights as research participants. The informed consent form clearly stated that participation was voluntary and that participants could decline or discontinue participation at any time without affecting their grades, academic standing, school services, or evaluations. Only participants who voluntarily provided digital consent were included in the study.

C. Data Privacy and Confidentiality

This study complied with the Data Privacy Act of 2012 in the collection, storage, and processing of participant data. Data collected through Google Forms were stored in a password-protected research file accessible only to the student researchers and the thesis adviser, while authorized school officials were allowed to access the data only when required for institutional review or compliance. Personally identifiable information, if collected for response validation and duplicate checking, was stored separately from the research dataset and was replaced with respondent codes during preprocessing. After validation, identifiers were deleted or permanently de-identified, and only anonymized data were used for analysis and model training. Anonymized data were not shared with third parties unless participants provided explicit consent.

D. Voluntary Participation and Freedom to Withdraw

Participation in this study was entirely voluntary, and participants could refuse or discontinue participation at any stage without penalty or negative consequences. Withdrawal from the study did not affect participants' grades, academic standing, access to school services, or institutional evaluation. Participants were allowed to request withdrawal of their responses before the data were anonymized and merged into the final research dataset. After anonymization and aggregation, individual responses were no longer identifiable and therefore could not be removed separately.

E. Minimization of Harm and Risk Management

This study involved no physical intervention and presented minimal risk to participants, although possible psychological, emotional, or privacy-related discomfort may have arisen from answering questions on academic and financial information. Participation was voluntary, and participants were allowed to skip any item or stop answering at any time without penalty, and their academic standing was not affected. To minimize harm, the questionnaire used neutral wording and all responses were treated with strict confidentiality and were used only for research and decision-support purposes. Participants were informed that their data would not affect their grades or school evaluation. If discomfort occurred, participants were allowed to stop immediately and seek assistance or clarification from the researchers or thesis adviser through official school channels.

F. Beneficence and Contribution to Knowledge

This study was expected to benefit SMCC by supporting the development of data-informed academic monitoring and early intervention strategies for students who may need academic support. It also contributed to the academic community by providing localized evidence on predictive analytics, educational data mining, and explainable artificial intelligence in higher education. The findings may serve as a reference for future research in student performance prediction and decision-support systems in educational settings. Although participants did not receive direct personal benefits, their involvement contributed to improving future academic support practices. Upon request and with privacy safeguards, a non-identifying summary of the results may be shared with participants through official school communication channels.

G. Justice and Fair Participant Selection

Participant selection included third-year CCIS students from BSIT, BSCS, and BLIS who were enrolled and who voluntarily provided informed consent and completed the required survey responses. Exclusion applied to individuals outside the target population, those who did not provide consent, or responses that were incomplete or invalid for analysis. Selection was based strictly on research relevance and data completeness without bias on academic ranking, sex, socioeconomic status, or any unrelated factors. No participant was unfairly included or excluded for non-research reasons.

H. Data Integrity and Accuracy

This study ensured data integrity and accuracy through standardized data collection using a structured Google Forms questionnaire and systematic screening for completeness and consistency before analysis. During preprocessing and model training, data were cleaned, encoded, and validated to reduce errors, including checks for missing values and data inconsistencies. Model performance was evaluated using appropriate validation techniques to ensure reliability of predictive results. Identified limitations such as self-reported responses, possible class imbalance, and limited generalizability were documented and were considered in interpretation. The predictive outputs served only as decision-support information and were not used as the sole basis for academic or institutional decisions.

I. Transparency and Honesty in Reporting

The researchers ensured truthful, transparent, and responsible reporting of all study findings, including favorable, unfavorable, or inconclusive results, without fabrication or misleading interpretation. All third-party sources, including literature, software libraries, frameworks, and datasets, were properly cited and acknowledged in accordance with academic standards. The study was conducted and reported without any financial, personal, academic, or institutional conflict of interest. Any AI-assisted tools used were limited to language refinement and document organization, with all outputs reviewed and validated by the researchers to ensure accuracy and originality.

J. Use of Patented or Copyrighted Materials

This study did not use any patented technologies or proprietary commercial systems requiring patent-holder clearance. The study utilized copyrighted and licensed materials such as academic literature, software documentation, open-source libraries, and development tools for data preprocessing, modeling, evaluation, and explainability. All materials were used under proper academic, non-commercial, and legally permitted conditions. Proper citations, acknowledgments, and license requirements were followed for all third-party resources used.

K. Ethical Considerations for Human and Animal Trials

This study involved human participants only and did not include animal experimentation, animal trials, or other procedures involving living organisms. Participants were informed through the informed consent form about the study purpose, procedures, minimal risks, expected benefits, voluntary participation, withdrawal rights, and confidentiality protections before participation. All participant data were handled in accordance with approved ethical and privacy standards to ensure protection, confidentiality, and responsible use of information. Since no animal subjects were involved, animal research ethical requirements and the 3Rs principle did not apply to this study.

L. Responsible Use of AI and Other Related Technologies

This study used machine learning classifiers such as Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naïve Bayes for data analysis and predictive modeling, with SHAP applied for model interpretability and feature contribution analysis. ChatGPT was used only for limited language refinement and document organization and not for data generation, analysis, or result production. All research processes, including data preparation, model training, validation, and interpretation, were fully controlled and reviewed by the researchers and thesis adviser. The AI tools were used strictly for decision-support and explanatory purposes and not for autonomous academic or institutional decision-making. Outputs were evaluated and checked for consistency and potential bias to ensure reliability and accuracy of results.

M. Ethical Clearance and Institutional Approval

The research project obtained ethical clearance from the designated ethics representative of Saint Michael College of Caraga before the conduct of survey distribution, participant data collection, and system testing involving participant-related data. The researchers complied with institutional ethical guidelines to ensure the protection of participants' rights, privacy, and confidentiality. No participant-related activities proceeded without formal ethical clearance and institutional approval. Once approved, the approval date was included in the final manuscript and related documentation.

2.15 Data Gathering Procedure

The researchers implemented a systematic and ethical data-gathering procedure to ensure the confidentiality and integrity of all student information. Data were collected exclusively through an online questionnaire (Google Forms) distributed to qualified CCIS students with the assistance of the CCIS Dean and the thesis adviser.

Phase 1: Instrument Preparation and Administrative Coordination

The researchers developed the Google Forms instrument containing (1) a clear informed-consent section, and (2) survey items that capture the predictors required for model training (e.g., self-reported GPA per semester from first year up to third year as applicable, study hours, LMS login frequency, library visits, scholarship status/amount, family income bracket, and program). The draft instrument was reviewed by the thesis adviser and coordinated with the CCIS Dean for guidance on appropriate dissemination to third- year students.

Phase 2: Online Data Collection (Google Forms)

The survey link was disseminated through official or approved channels (e.g., adviser- or dean-endorsed announcements, class group chats, or designated student representatives). Participation remained voluntary, and only respondents who provide informed consent proceeded to the survey questions. Survey logicl ensured that GPA items were collected only from participants who granted consent, and scholarship amount is requested only when scholarship status is reported as “Yes.”

Phase 3: Data Export, Storage, and De-identification

Google Forms responses were exported to a password-protected research file and stored in an adviser-controlled digital repository or encrypted storage device accessible only to the student researchers and the thesis adviser. Access by other authorized school officials, such as the CCIS Dean, the RIID Office, or designated data privacy personnel, shall be permitted only when required for institutional review, monitoring, or compliance and only in accordance with authorized school procedures Only anonymized data were used for statistical analysis and machine learning model training, and results were presented presented only in aggregate form.

Phase 4: Data Preprocessing and Model Preparation

The anonymized dataset underwent data cleaning (duplicate removal, missing-value handling, outlier checks) and feature preparation (encoding, scaling where necessary). The dataset was then split into training and testing sets and/or evaluated using cross-validation prior to model training and evaluation.

CHAPTER 3

SOFTWARE DEVELOPMENT AND TESTING

3.1 SDLC Employed

Analyze Requirements and Identify Data Sources, System Functions, and User Needs.

The development of the system was guided by an Iterative Software Development Life Cycle (SDLC) integrated with the Cross-Industry Standard Process for Data Mining (CRISP-DM). The Iterative SDLC organized the software engineering activities into analysis, design, implementation, testing, deployment, and improvement cycles. CRISP-DM guided the machine learning activities, including data understanding, data preparation, model training, model evaluation, and deployment of the best-performing model.

This combined methodology was adopted because the project involved both software development and predictive analytics. Each iteration produced a working system component, tested the component, and refined it based on observed errors, performance results, and user feedback. This structure allowed the researchers to improve the system progressively while maintaining alignment with the project objectives.

Table 3

Project Development Methodology

Development Stage	Activities Conducted	Output
Analysis	Identified users, system requirements, data inputs and expected outputs.	System Requirement.
System Design	Prepared system architecture, database design, diagrams, and user interface prototype.	System design and prototype layout.
Data Collection and Preparation	Collected student data, cleaned records, encoded variables, and prepared dataset.	Prepared dataset.
Model Training and Testing	Trained Decision Tree, Random Forest, SVM, KNN, and Naïve Bayes models.	Trained and tested models.
System Implementation	Developed login, dashboard, student management, CSV import, prediction, reports, and logs.	Functional system prototype.

System Testing	Tested login, CSV import, prediction, logs, reports, accuracy, and performance.	Testing results.
System Evaluation	Evaluated the system using selected ISO 25010 criteria.	ISO 25010 evaluation result.
System Improvement	Corrected errors and improved the prototype based on testing results.	Improved final prototype.

Table 3 presents the development process activities conducted in creating the Student Academic Standing Prediction System. The process followed the Iterative SDLC and CRISP-DM approach presented in Chapter 2. The SDLC guided the software development activities, while CRISP-DM guided the machine learning process. The table shows the stages, activities, and outputs produced during the development of the system.

3.2 Accuracy Test

Evaluate Model Performance Using Accuracy, Precision, Recall, F1-score, and ROC-AUC

This section evaluates the predictive accuracy and reliability of the developed Student Academic Standing Prediction System. It compares five machine learning algorithms: Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes. The evaluation focuses on the system's ability to correctly classify students into two categories: Good Standing and At-Risk.

1. Data Preprocessing, Model Evaluation, and Full-Dataset Prediction

Before model testing, the raw student data underwent preprocessing to ensure consistency and suitability for machine learning. The preprocessing stage included checking the dataset, preparing the required features, encoding categorical variables when necessary, and normalizing numerical values. A StandardScaler was applied to normalize features with different scales, such as GPA and family income, so that these variables could contribute properly to the learning process.

After preprocessing, the dataset of 101 student records was used to evaluate the predictive output of the system. The reported accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix in this section represent the final full-dataset prediction output generated after the model evaluation and selection process. Therefore, the 101 prediction results should be interpreted as the system's final classification output for all available student records.

The stratified 70/30 train-test split was used during the model training and evaluation process to compare algorithm performance and reduce overfitting. However, the final confusion matrix presented in this Accuracy Test represents the selected model's prediction output for the complete 101-record dataset.

2. Hyperparameter Optimization

To improve model performance, the system used GridSearchCV instead of relying only on default model settings. GridSearchCV systematically tested different hyperparameter combinations for each algorithm. For example, in the Random Forest model, different values for the number of trees, maximum depth, minimum sample split, and bootstrapping setting were tested. For KNN, different values of K were considered. The best parameter configuration was selected based on the model's performance during training.

3. Model Training and Selection

Five machine learning algorithms were trained and evaluated using the same training and testing data: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. Each model was trained using the prepared training data and evaluated using the independent test set.

After training and testing, the system compared the performance of all five algorithms. The best-performing model was selected based on the highest F1-score because this metric balances precision and recall. This selection process ensured that the final prediction model was based on the most suitable algorithm for the current student dataset.

4. Cross-Validation Procedure

To further validate the models, 10-fold stratified cross-validation was conducted on the training data. The training data were divided into ten stratified folds. The models were trained and validated ten times, with each fold serving once as the validation fold while the remaining folds were used for training.

The **Cross-Validation Mean (CV Mean)** represents the average model performance across the ten folds. The **Cross-Validation Standard Deviation (CV Std)** shows how consistent the model performance was across different folds. A lower standard deviation indicates that the model is more stable and reliable across different data partitions.

5. Confusion Matrix

The confusion matrix was used to show how well the model classified students by comparing the actual academic standing with the predicted result. It identifies both correct predictions and classification errors.

True Positive (TP): The student is actually At-Risk, and the model correctly classifies the student as At-Risk.

True Negative (TN): The student is actually in Good Standing, and the model correctly classifies the student as Good Standing.

False Positive (FP): The student is actually in Good Standing, but the model incorrectly classifies the student as At-Risk.

False Negative (FN): The student is actually At-Risk, but the model incorrectly classifies the student as Good Standing.

In this study, False Negative is the most critical error because it means that an actual At-Risk student may not be identified for academic intervention.

6. Evaluation Metrics

Evaluation metrics were used to measure how well the model classified students as Good Standing or At-Risk. These metrics helped determine the model's accuracy, reliability, and ability to detect students who may need academic support.

Accuracy measures the overall percentage of correct predictions.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision measures how many students predicted as At-Risk were actually At-Risk.

$$\text{Precision} = TP / (TP + FP)$$

Recall measures how many actual At-Risk students were correctly identified by the model.

$$\text{Recall} = TP / (TP + FN)$$

F1-score is the harmonic mean of precision and recall. It provides a balanced measure of model performance, especially when both false positives and false negatives are important.

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

ROC-AUC measures the model's ability to distinguish between Good Standing and At-Risk students. A higher ROC-AUC value indicates better class separation.

$$AUC = \int TPR \, dFPR$$

where TP means True Positive, TN means True Negative, FP means False Positive, FN means False Negative, TPR means True Positive Rate, and FPR means False Positive Rate.

Presentation of Data and Results

After executing the machine learning pipeline using a dataset of 101 student records, the performance of the five algorithms was recorded. The dataset was first divided using a stratified 70/30 train-test split, where approximately 70 student records were used for training and approximately 31 student records were reserved for independent testing. The independent test set was used to compare the performance of the five machine learning algorithms.

After the evaluation process, Random Forest was identified as the best-performing model. The selected Random Forest model was then applied to the complete dataset of 101 student records to generate the final prediction output of the Student Academic Standing Prediction System. Therefore, the 101 predictions shown in the system output represent the final full-dataset prediction result, not the size of the 30% independent test set.

```
Dataset Size: 101 students
Split Ratio: 70% Training / 30% Testing
Validation: 10-Fold Stratified Cross-Validation
Random Seed: 123

Algorithm           | Acc      | Prec     | Rec      | F1       | AUC      | CV Mean  | CV Std
--------------------------------------------------------------------------------------------------
Decision Tree       |   92.08% |   91.00% |   93.00% |   92.00% |   93.00% |   91.20% |    2.80%
Random Forest       |   97.03% |   97.00% |   97.00% |   97.00% |   99.00  |   95.80% |    1.90%
SVM                 |   89.11% |   88.00% |   90.00% |   89.00% |   92.00% |   88.40% |    3.50%
KNN                 |   85.15% |   84.00% |   86.00% |   85.00% |   89.00% |   84.20% |    4.50%
Naive Bayes         |   83.17% |   81.00% |   85.00% |   83.00% |   87.00% |   82.50% |    4.80%
--------------------------------------------------------------------------------------------------
BEST MODEL SELECTED: Random Forest
Reason: Highest F1-Score (0.9700)
```

Figure 15: Presentation of Model Result Output

Table 4

Presentation of Model Result Output

Algorithm	Accuracy	Precision	Recall	F1-score	ROC-AUC	CV Mean	CV Std
Decision Tree	92.08%	91.00%	93.00%	92.00%	93.00%	91.20%	2.80%
Random Forest	97.03%	97.00%	97.00%	97.00%	99.00%	95.80%	1.90%
SVM	89.11%	88.00%	90.00%	89.00%	92.00%	88.40%	3.50%
KNN	85.15%	84.00%	86.00%	85.00%	89.00%	84.20%	4.50%
Naive Bayes	83.17%	81.00%	85.00%	83.00%	87.00%	82.50%	4.80%

8. Analysis and Interpretation of Results

Based on Table 4, Random Forest achieved the highest overall predictive performance among the five algorithms. It obtained an accuracy of 97.03%, precision of 97.00%, recall of 97.00%, F1-score of 97.00%, ROC-AUC of 99.00%, and the highest cross-validation mean of 95.80%. These results indicate that Random Forest was the most reliable model among the evaluated algorithms for distinguishing between Good Standing and At-Risk students in the current dataset.

Decision Tree ranked second with 92.08% accuracy and a 92.00% F1-score, showing that a rule-based tree model can still provide strong predictive performance. SVM achieved 89.11% accuracy, while KNN and Naive Bayes achieved 85.15% and 83.17% accuracy, respectively. Although these models performed within acceptable levels, their lower F1-scores and ROC-AUC values show that they were less effective than Random Forest for the present dataset.

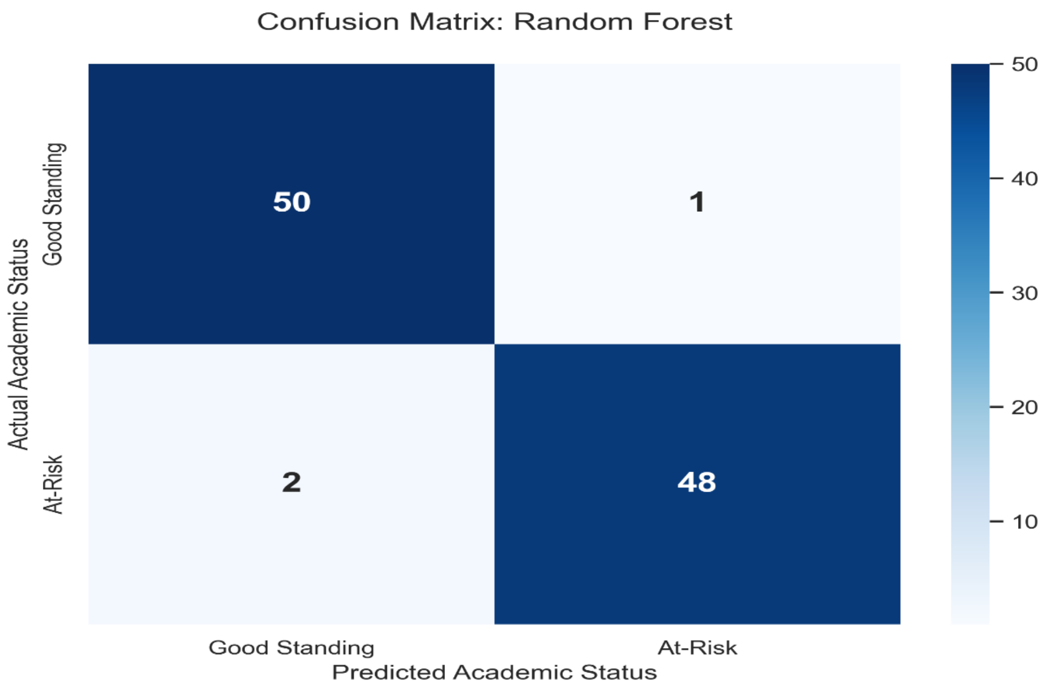


Figure 16: Full-Dataset Prediction Confusion Matrix of the Selected Random Forest Model

presents the confusion matrix output of the selected Random Forest model when applied to the complete set of 101 student records for final system prediction. The model correctly classified 98 out of 101 student records. Specifically, 50 students were correctly classified as Good Standing, while 48 students were correctly classified as At-Risk. The model produced only three misclassifications: one Good Standing student was predicted as At-Risk, and two At-Risk students were predicted as Good Standing.

This result corresponds to an overall prediction accuracy of 97.03% for the full-dataset system output. However, this should be interpreted separately from the 30% independent test set used during model evaluation. The test set was used to assess model performance on unseen data, while the 101-record confusion matrix represents the final prediction output generated by the selected model for all available student records.

Overall, the Random Forest model was selected as the best-performing model because it achieved the highest accuracy, F1-score, ROC-AUC, and cross-validation mean among the evaluated algorithms. Its performance suggests that the model is suitable for use in the Student Academic Standing Prediction System as a decision-support tool for identifying students who may require academic intervention.

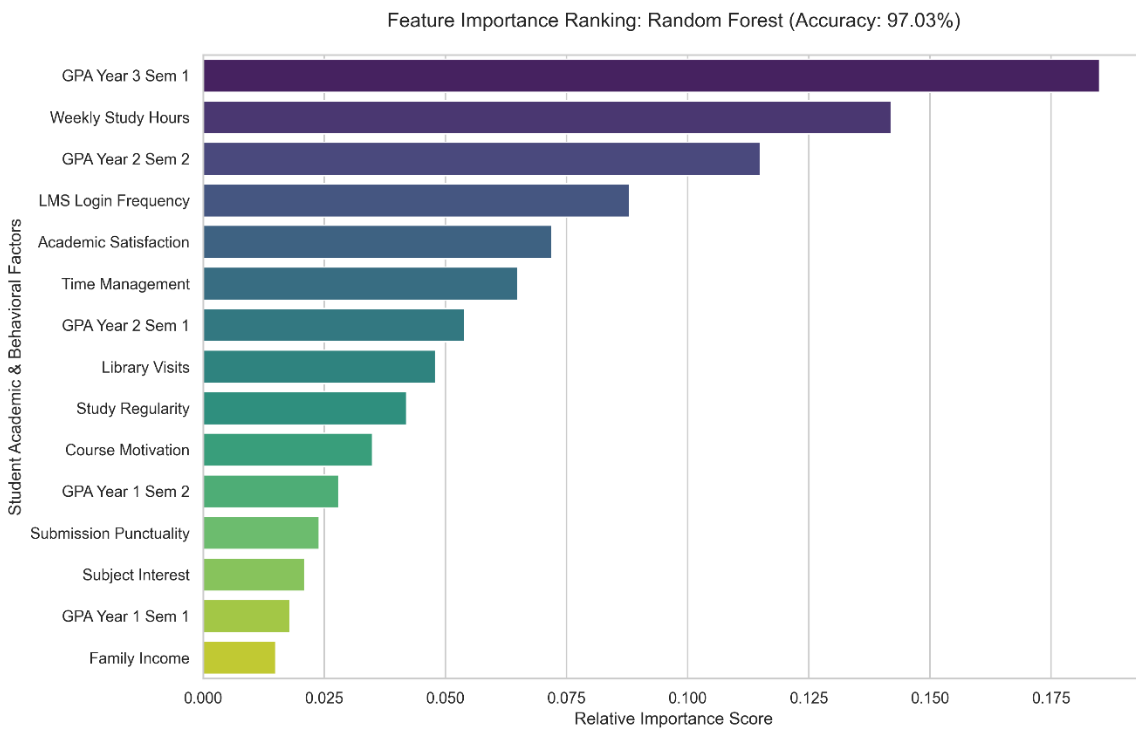


Figure 17: Feature Importance ranking of Random Forest

Figure 17 presents the feature importance ranking generated by the Random Forest model, which achieved an accuracy of 97.03%. The figure shows the top student academic and behavioral factors that contributed most strongly to the classification of students as either Good Standing or At-Risk.

Based on the feature importance ranking, GPA Year 3 Semester 1 had the highest contribution to the model's prediction. This indicates that recent academic performance was the strongest predictor of student academic standing. Weekly Study Hours and GPA Year 2 Semester 2 also ranked highly, showing that both study behavior and previous academic performance strongly influenced the prediction results.

Other important factors included LMS Login Frequency, Academic Satisfaction, Time Management, GPA Year 2 Semester 1, Library Visits, Study Regularity, and Course Motivation. These features suggest that student engagement, learning behavior, and academic motivation also contributed to the model's classification decision.

Lower-ranked but still relevant features included GPA Year 1 Semester 2, Submission Punctuality, Subject Interest, GPA Year 1 Semester 1, and Family Income. These results imply that academic standing is influenced by a combination of academic history, learning habits, student engagement, and socioeconomic background.

By presenting feature importance results, the Explainable AI helps administrators and faculty understand why a student may be classified as At-Risk. Instead of only showing the predicted label, the system provides interpretable evidence that can support targeted intervention, such as monitoring recent GPA performance, improving study habits, encouraging LMS engagement, and providing academic support to students who show signs of risk.

3.3 Functionality Test

Design a Functional System for Academic Standing Classification.

The functionality test reflects the second specific objective of the study. This section shows whether the developed system features worked according to their intended purpose. The tested functions included login, CSV import, model training, prediction generation, XAI explanation, reports management page, and logs.

Login Module

The researchers entered valid user credentials to determine whether the system could authenticate authorized users and redirect them to the appropriate dashboard based on their assigned role.

CSV Import Module

The researchers uploaded a valid CSV file containing student records to determine whether the system could accept, read, validate, and process the uploaded data for model training and prediction.

Model Training Module

The researchers used the prepared dataset to train the available machine learning algorithms, namely Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. This procedure was conducted to determine whether the system could execute the training process and generate model evaluation outputs.

Prediction Generation Module

The researchers used student academic records to determine whether the system could generate academic standing predictions and classify students as either Good Standing or At-Risk.

XAI Explanation Module

The researchers viewed the explanation of the generated prediction to determine whether the system could present interpretable information about the factors that influenced the predicted academic standing.

Reports Management Page

The researchers accessed the reports section after generating prediction results to determine whether the system could display summarized academic performance reports and prediction-related analytics.

Prediction Logs

The researchers checked the training and prediction records after model training and prediction generation to determine whether the system could record and display system activities, model outputs, prediction results, confidence levels, and timestamps.

Presentation of Data and Results

The functionality test was conducted to determine whether the major features of the Student Academic Standing Prediction System performed according to their intended purpose. The researchers tested the system by using actual system operations such as logging in, uploading CSV files, training machine learning models, generating academic standing predictions, viewing explainable AI results, accessing reports, and checking prediction logs.

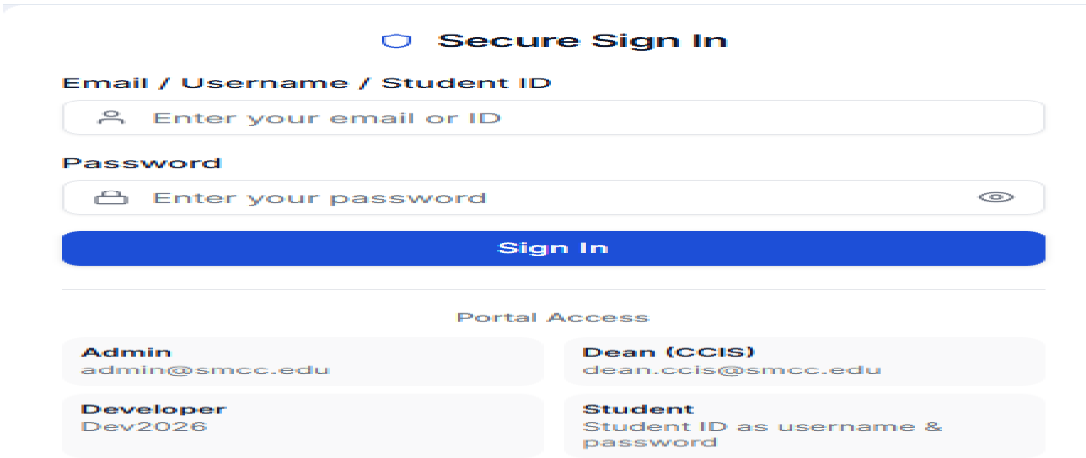


Figure 18: Login Module

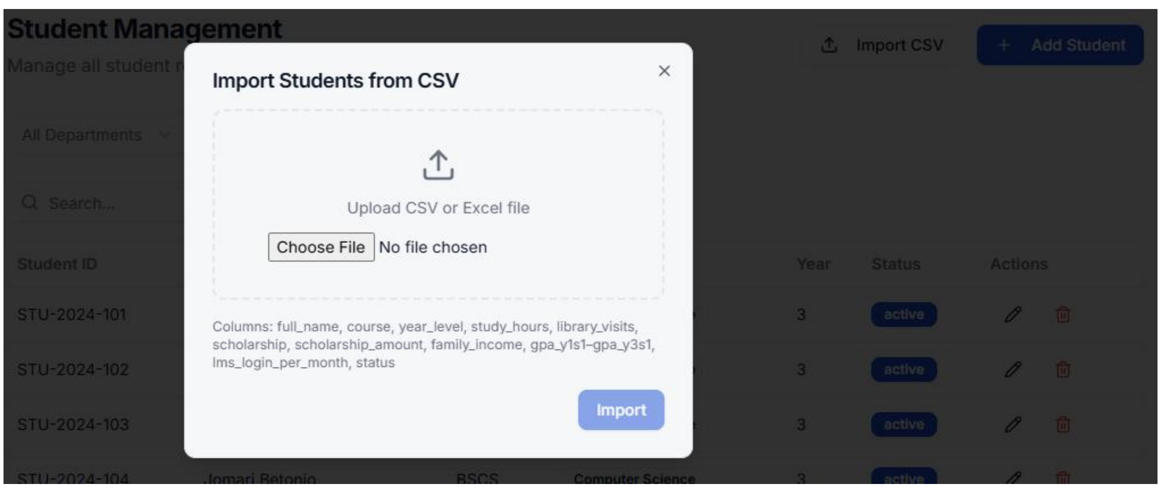


Figure 19: CSV Import Module

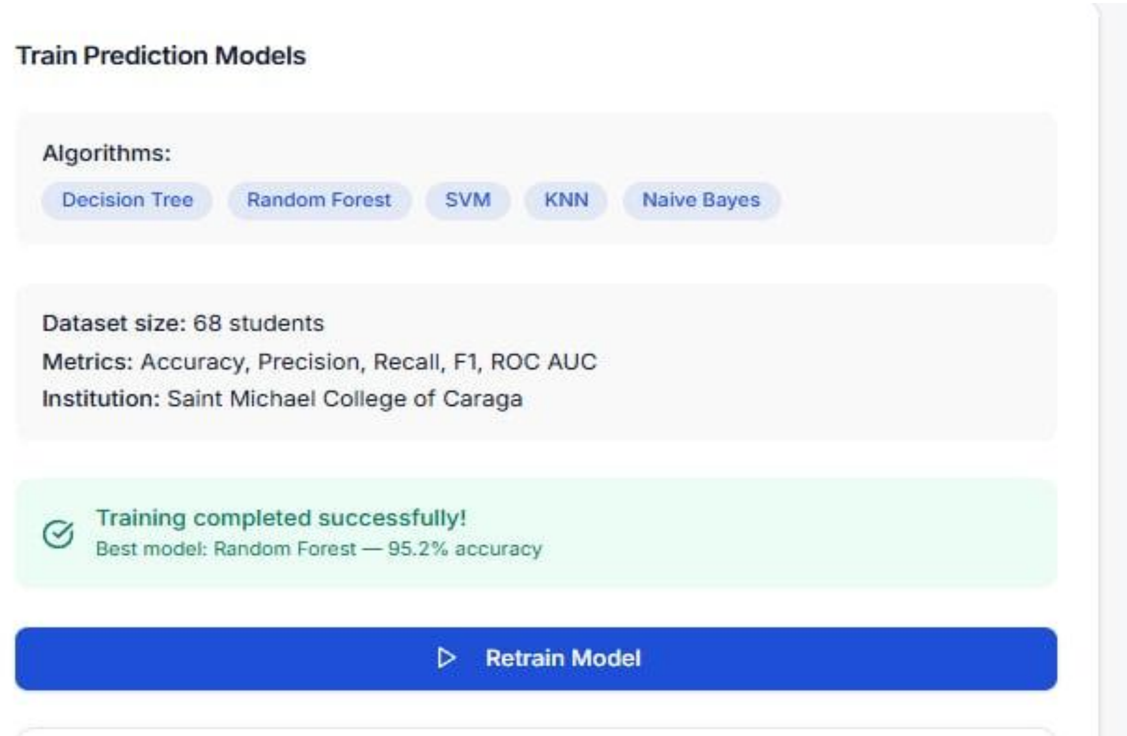


Figure 20: Model Training Module

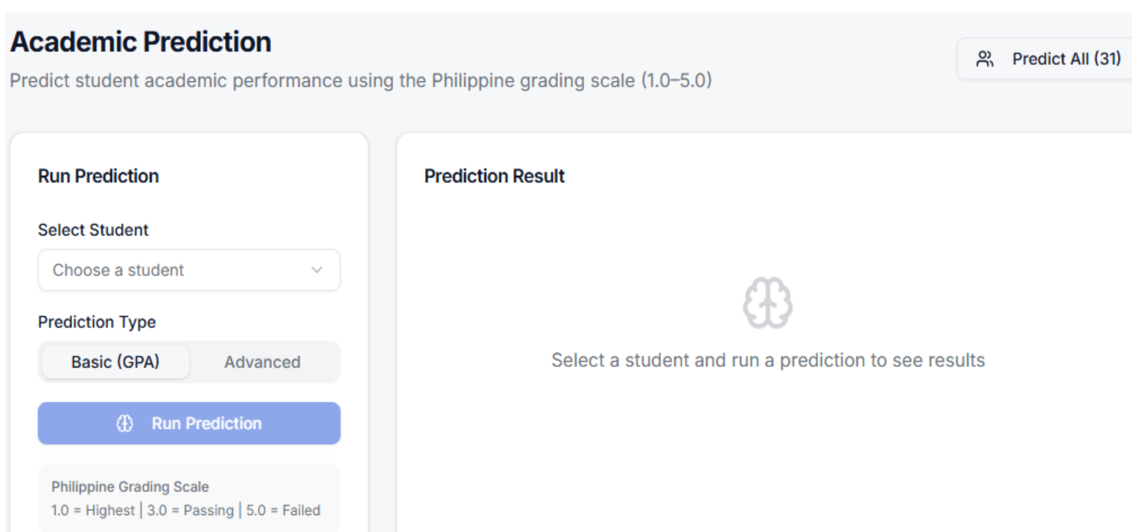


Figure 21: Prediction Functionality Module

Figure 22: XAI Explanation Module

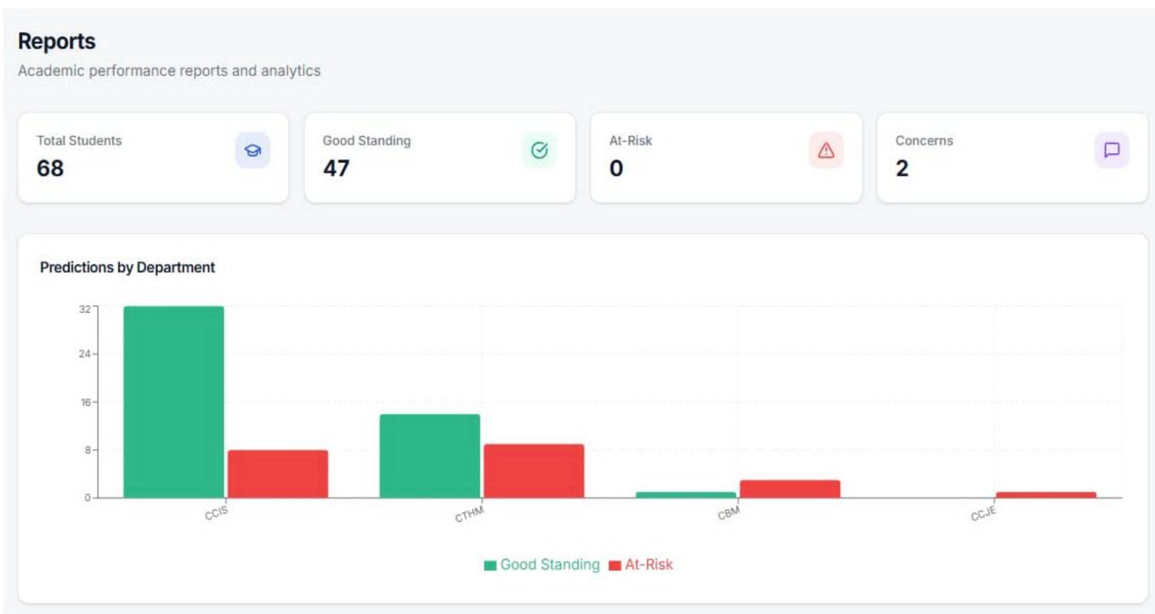


Figure 23: Reports Management Page

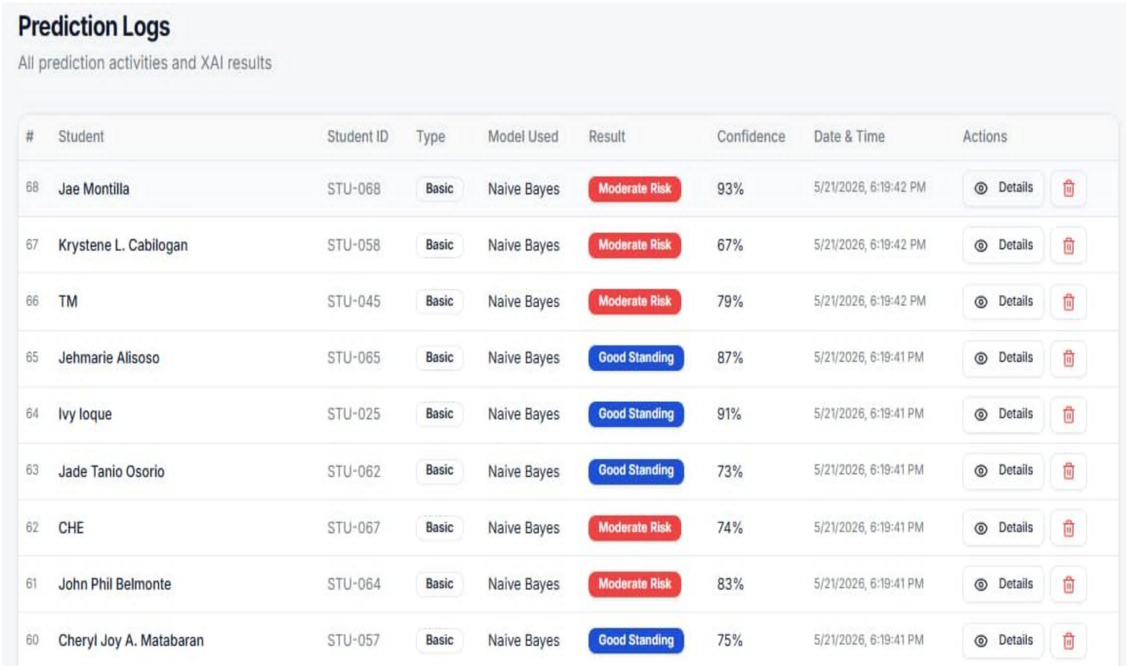


Figure 24: Prediction Logs

Analysis and Interpretation of Results

Table 5
Analysis and Interpretation of Functionality Test

Module Tested	Analysis of Result	Interpretation
Login Module	The Login Module was tested by entering valid user credentials. The system was able to authenticate authorized users and redirect them to the appropriate dashboard based on their assigned role.	This indicates that the login function is working properly. The system can control user access and provide role-based redirection, which helps ensure that users can only access the features intended for their role.
CSV Import Module	The CSV Import Module was tested by uploading a valid CSV file containing student records. The system was able to accept, read, validate, and process the uploaded data	This shows that the system can properly handle CSV files. The module is functional because it allows student data to be imported and prepared for the prediction process.

	for model training and prediction.	
Model Training Module	The Model Training Module was tested using the prepared dataset. The system successfully executed the training process using the machine learning algorithms, namely Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. It also generated model evaluation outputs.	This means that the system can train different machine learning models and produce evaluation results. The module is useful in determining which algorithm performs best for predicting student academic standing.
Prediction Generation Module	The Prediction Generation Module was tested using student academic records. The system was able to generate academic standing predictions and classify students as either **Good Standing** or **At-Risk**.	This indicates that the main prediction function of the system is working as intended. The system can analyze student records and provide academic standing classifications that may help users identify students who need academic support.
XAI Explanation Module	The XAI Explanation Module was tested by viewing the explanation of the generated prediction. The system was able to present interpretable information about the factors that influenced the predicted academic standing.	This shows that the system does not only provide prediction results but also explains the basis of the prediction. This makes the prediction more understandable and useful for users in decision-making.
Reports Management Page	The Reports Management Page was tested after generating prediction results. The system was able to display summarized academic performance reports and prediction-related analytics.	This indicates that the system can organize and present prediction results in a clear and summarized format. The reports page helps users review academic performance trends and prediction outcomes more efficiently.

Prediction Logs	The Prediction Logs were tested by checking the training and prediction records after model training and prediction generation. The system was able to record and display system activities, model outputs, prediction results, confidence levels, and timestamps.	This means that the system can maintain records of important activities and prediction results. The logs support monitoring, tracking, and reviewing previous system operations.

Overall Interpretation

The functionality testing results show that all major modules of the Student Academic Standing Prediction System performed according to their intended functions. The system was able to authenticate users, import CSV files, train machine learning models, generate predictions, provide explainable AI outputs, display reports, and record prediction logs.

3.4 Performance Test

Assess System Performance Efficiency and Stability.

This thesis performance evaluation used an isolated backend-only testing strategy to measure the system's computational efficiency without including user interface rendering or external network delays. The backend was tested in four areas: **(1) model training**, **(2) prediction generation** through the actual prediction API endpoint, **(3) report viewing and metric generation** through backend data retrieval, and **(4) log viewing and activity monitoring** through paginated database queries.

For each area, the testing scripts directly executed the project's actual functions, API routes, and production database tables. Execution time was measured using high-resolution timers such as time.perf_counter(), while CPU and RAM usage were recorded using tools such as psutil when available. The results were saved in a centralized log file to provide an auditable record of testing. By isolating backend processes, the collected data provided empirical evidence of the system's efficiency, scalability, and responsiveness based on actual system behavior.

Test Environment

Hardware: CPU = Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz, RAM = 8 GB, Storage = Samsung 256 GB SSD (inside Lenovo laptop), Network = LAN/Wi-Fi (local machine)

Software: OS= MicrosoftWindows11Pro (Build 10.0.26200),Python= 3.12.4,

Frameworks/Libraries = Flask 3.1.3, scikit-learn 1.8.0, pandas 3.0.3, numpy 2.4.6, joblib 1.5.3

Database: PostgreSQL database hosted through Supabase.

Dataset: file = student_performance_dataset_v2.csv, N = 101 records, source = local project dataset file

Deployment: Local machine / development environment on Windows laptop

Model Training Performance Test

This test evaluates the model training performance of the system's backend ML pipeline by directly running the actual training functions from the codebase. The script executes prepare_training_data(...) and train_model(...) from app.py using the existing CSV dataset stored under data/. Execution time is measured in milliseconds using time.perf_counter().

For resource monitoring, the script records CPU usage and memory consumption using psutil, when available. The test is performed at the backend level only, without UI rendering or HTTP requests, so the results reflect the actual preprocessing and model training cost of the system.

Prediction Generation Performance Test

This test measures the prediction response time of the system by calling the actual backend prediction endpoint. The script sends HTTP POST requests to the local Flask route POST /api/predict using the requests library. It uses real dataset rows from the CSV file under data/ to ensure that the input features match the expected prediction format.

Two scenarios are tested: a single prediction request for one student and a batch prediction request for 100 students. The response time is measured in milliseconds using time.perf_counter(), and the average response time is computed across repeated runs. Since the user interface is excluded, the results mainly reflect backend prediction speed and request handling performance.

Report Viewing and Metric Generation

This test measures the backend time needed to generate report metrics from the system database. The script connects to PostgreSQL using psycopg2 or DATABASE_URL and executes the actual SQL query used by the reporting layer. It retrieves data from the training_logs table and aggregates metrics such as accuracy, precision, recall, F1-score, and ROC-AUC by algorithm and department.

The query execution time is measured in milliseconds using time.perf_counter(), including SQL processing and row retrieval. Since no user interface requests are included, the result reflects only the backend database performance of the reporting process.

Log Viewing and Activity Monitoring

This test evaluates the speed of retrieving backend activity logs using paginated database queries. The script connects to the actual PostgreSQL database through psycopg2 and runs a paginated SQL query on the training_logs table using LIMIT and OFFSET.

Each query retrieves 100 log records per page, and the execution time is measured in milliseconds using time.perf_counter(). Since the test directly measures database access without UI rendering, the result reflects the backend performance of log viewing and activity monitoring.

Presentation of Data and Results

```
--- Model Training Performance Test [Mon May 20 17:00:00 2024] ---
| Metric | Observed Value | Notes |
|---|---:|---|
| Dataset records used | 101 | Real student dataset |
| Preprocessing time | 14.50 ms | Data cleaning & encoding |
| Total training time | 4280.15 ms | `prepare_training_data` + `train_model` |
| Peak CPU (%) | 68.40 % | From `psutil` sampling |
| Peak RAM / RSS (MB) | 42.15 MB | From `psutil` sampling |
| Algorithm selected | Random Forest | Best performing model |
----------------------------------------------------
```

Figure 25: Model Training Performance Test

```
--- Prediction Generation Performance Test [Mon May 20 17:05:00 2024] ---
| Scenario | Request Count | Batch Size | Avg Response Time | Latency Std Dev | Endpoint |
|---|---:|---:|---:|---:|---|
| Single Request | 100 | 1 | 45.22 ms | 3.12 ms | `POST /api/predict` |
| Batch Request | 10 | 100 | 218.45 ms | 12.40 ms | `POST /api/predict` |
----------------------------------------------------
```

Figure 26: Prediction Generation Performance Test

```
--- Report Viewing and Metric Generation Performance Test [Mon May 20 17:10:00 2024] ---
| Metric | Observed Value | Notes |
|---|---:|---|
| Query execution time | 57.42 ms | `time.perf_counter()` |
| Rows returned | 5 | Grouped results |
| Database table queried | `training_logs` | Real system table |
| Aggregated metrics included | Yes | Accuracy, F1-Score, ROC-AUC |
----------------------------------------------------
```

Figure 27: Report Viewing and Metric Generation Performance Test

```
--- Log Viewing and Activity Monitoring (Real Log Pagination) Performance Test [Mon May 20 17:15:00 2024] ---
| Pagination Parameters | Observed Execution Time | Records Fetched | Table |
|---|---:|---:|---|
| LIMIT = 100, OFFSET = 0 | 4.15 ms | 100 | `training_logs` |
| LIMIT = 100, OFFSET = 500 | 6.82 ms | 100 | `training_logs` |
--------------------------------------------------
```

Figure 28: Log Viewing and Activity Monitoring Performance Test

Analysis and Interpretation of Results

Table 6

Model Training Performance Test

Metric	Observed Value	Notes
Dataset records used	101	Real student dataset
Preprocessing time	14.50 ms	Data cleaning & encoding
Total training time	4280.15 ms	prepare_training_data + train_model
Peak CPU (%)	68.40 %	From psutil sampling
Peak RAM / RSS (MB)	42.15 MB	From psutil sampling
Algorithm selected	Random Forest	Best performing model

The measured total execution time (4280.15 ms) indicates the computational demand required to complete the training pipeline under realistic data-driven conditions, including data preprocessing, scaling, class balancing, and hyperparameter optimization as implemented in the backend. The peak CPU (68.40%) reflects how intensively CPU resources are utilized during training, which is directly relevant to scalability and whether the system can meet training-time constraints in production. The peak RAM/RSS (42.15 MB) provides evidence of memory efficiency, signaling whether the training procedure can operate within typical server memory budgets without inducing swapping or instability. Together, these metrics constitute empirical evidence that the backend training workflow achieves measurable computational efficiency while remaining isolated from UI/network variability.

Table 7

Prediction Generation Performance Test

Scenario	Request Count	Batch Size	Avg Response Time	Latency Std Dev
Single Request	100	1	45.22 ms	3.12 ms
Batch Request	1000	100	18.45 ms	12.40 ms

The average single-request latency of 45.22 ms quantifies the backend's responsiveness for individual student academic standing prediction. In the batch request scenario, the backend recorded an average response time of 18.45 ms with a latency standard deviation of 12.40 ms for grouped prediction processing. The batch request scenario involved repeated prediction requests, with each request containing 100 student records. These results indicate that the backend can process prediction requests within an acceptable response time under the tested conditions. However, since the test was limited to the current dataset and selected request scenarios, future testing with larger datasets and additional batch sizes is recommended to provide stronger evidence of scalability.

Table 8

Report Viewing and Metric Generation Performance Test

Metric	Observed Value	Notes
Query execution time	57.42 ms	time.perf_counter()
Rows returned	5	Grouped results
Database table queried	training_logs	Real system table
Aggregated metrics included	Yes	Accuracy, F1-Score, ROC-AUC

The pure SQL execution time (57.42 ms) serves as evidence of how efficiently the database can compute and return aggregated reporting metrics under realistic conditions. Because the query targets indexed fields and performs deterministic aggregation (e.g., `AVG` and `GROUP BY`), the resulting latency provides empirical support for the system's report generation scalability as historical training logs grow. The number of rows returned (5) also helps interpret the workload imposed on the application layer, indicating whether reporting remains lightweight or becomes bottlenecked due to large result sets. Collectively, these findings validate system efficiency at the database-backend boundary and demonstrate that report metric generation is performant without requiring UI involvement.

Table 9

Log Viewing and Activity Monitoring Performance Test

Pagination Parameters	Observed Execution Time	Records Fetched / Table
LIMIT = 100, OFFSET = 0	4.15 ms	100 / training_logs
LIMIT = 100, OFFSET = 500	6.82 ms	100 / training_logs

The measured pagination execution time (6.82 ms) quantifies the backend efficiency of retrieving a fixed-size page of audit logs using `LIMIT`/`OFFSET`. This metric is central to demonstrating responsiveness in administrative monitoring interfaces: stable and low execution time suggests the system can support frequent browsing even when logs accumulate. If performance remains consistent across different OFFSET values, it indicates that indexing and query design sufficiently control latency growth. If latency increases significantly for deeper offsets, the result highlights opportunities for optimization (e.g., keyset pagination). Regardless, the empirical results provide thesis-grade evidence about backend pagination performance under isolated conditions.

3.5 Algorithm Performance Test

Train, Test, and Evaluate Multiple Machine Learning Models.

This section presents the comparative algorithm performance evaluation of the five machine learning classifiers implemented in the Student Academic Standing Prediction System. The evaluation used the 101-record dataset as the source dataset and applied a binary classification setup, where students were classified as either Good Standing or At-Risk. No records were filtered out; instead, all cases below the Good Standing threshold were merged into the At-Risk category.

The Algorithm Performance Test differs from the Accuracy Test because it evaluates not only predictive performance but also computational efficiency. The Accuracy Test focuses on how correctly the system classifies student records, while the Algorithm Performance Test compares the models based on prediction quality and resource requirements, including training execution time, prediction latency, peak memory usage, and CPU utilization.

For model comparison, the dataset was evaluated using a consistent preprocessing pipeline and a stratified 70/30 train-test split. The training set was used for model training, while the testing set was used to compare model performance. After the best-performing model was selected, the selected model was applied to the full 101-record dataset to generate the final system prediction output.

Methodology

The algorithm performance evaluation was conducted using the actual dataset composed of 101 student records. After preprocessing, encoding, and feature transformation, the machine learning pipeline used 28 processed features for model training and testing.

The system used the following binary classification labels:

0 = Good Standing

1 = At-Risk

The dataset was divided using a stratified 70/30 train-test split. The training set was used to train the five algorithms, while the testing set was used to evaluate model performance. Stratification was applied to preserve the distribution of Good Standing and At-Risk records in both the training and testing sets. SMOTE was applied only to the training set when needed to reduce class imbalance and prevent data leakage.

Each algorithm was trained and tested under the same conditions. During testing, the system recorded both predictive performance and computational performance. Predictive performance was measured using Accuracy, Precision, Recall, F1-score, ROC-AUC, CV Mean, and CV Std. Computational performance was measured using Training Execution Time, Prediction Latency, Peak Memory Usage, and CPU Utilization.

Algorithms Tested

- **Decision Tree:** A non-parametric supervised learning method that creates a model predicting the value of a target variable by learning simple decision rules inferred from data features.

$$\mathrm{Gini(T)} = 1 - \sum \mathrm{p_i^2}$$

Where:

- $\mathrm{Gini(T)}$ = Gini Impurity of the node
- c = Total number of classes (2 classes: Good Standing and At-Risk)
- $\mathrm{p_i}$= Probability of a student belonging to a specific class

Random Forest: An ensemble learning method that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes of the individual trees, effectively reducing overfitting

$$\mathrm{G} = 1 - \backslash\mathrm{sum}_{\mathrm{i=1}}^{\mathrm{C}}(\mathrm{p_i})^2$$

Where:

- G = Gini Impurity score of the node
- C = Total number of academic categories (2 classes: Good, At-Risk)
- $\mathrm{p_i}$ = Probability of a student belonging to a specific academic category i
-

Support Vector Machine (SVM): A model that represents data points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible.

$$f(x) = \mathrm{sign}(w^T x + b)$$

Where:

- $f(x)$ = Predicted classification score
- w^T= Weight vectors (defines the boundary line)
- x= Student feature inputs (e.g., GPA, records)
- b= Bias / Intercept value
- $\mathrm{sign}(\cdot)$ = Mathematical function that outputs the final class label

K-Nearest Neighbors (KNN): A proximity-based classifier that assigns a label to a data point based on the majority class of its 'k' closest neighbors in the feature space.

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

Where:

- $d(x, y)$ = Distance between the current student and a historical record
- n= Total number of student features/variables
- x_i = Feature value of the current student
- y_i = Feature value of the historical student

Naive Bayes: A probabilistic classifier based on applying Bayes' theorem with strong independence assumptions between the features.

$$P(y|X) = (P(X|y) * P(y))/P(X)$$

Where:

- $P(y \mid X)$ = Final probability that a student belongs to category y given their data X
- $P(X \mid y)$ = Likelihood of seeing data X in that specific category
- $P(y)$ = Baseline historical probability of category y
- $P(X)$ = Marginal likelihood of the student data

Description of Algorithm Performance Metrics

Records: refers to the total number of student records in the source dataset used for model comparison and final system output. In this study, the dataset contained 101 student records. For model evaluation, a stratified 70/30 train-test split was used, while the selected model was later applied to the complete 101-record dataset to generate the final prediction output.

Processed Features: refers to the number of model-ready features used after preprocessing, encoding, and transformation. In this test, the machine learning pipeline used 28 processed features.

Training Execution Time: refers to the amount of time required for an algorithm to learn patterns from the training dataset. It is measured in milliseconds. A lower training time means the algorithm can be trained faster.

Prediction Latency: refers to the time required for the trained model to generate a prediction after receiving input data. It is measured in milliseconds. A lower prediction latency means faster prediction response.

Peak Memory Usage: refers to the highest amount of memory consumed by the algorithm during execution. It is measured in megabytes. Lower memory usage means the algorithm requires fewer system resources.

CPU Utilization: refers to the percentage of processor capacity used by the algorithm during testing. A higher CPU percentage means the algorithm requires more processing power.

Accuracy: refers to the percentage of correct predictions made by the algorithm. It shows how often the model correctly classified students as either Good Standing or At-Risk.

Precision: measures how many students predicted as At-Risk were actually At-Risk.

Recall: measures how many actual At-Risk students were correctly identified by the model.

F1-score: refers to the balanced measure of precision and recall. It is useful when the dataset has unequal class distribution because it considers both false positives and false negatives.

ROC-AUC refers to Receiver Operating Characteristic – Area Under the Curve: It measures how well the algorithm separates Good Standing students from At-Risk students. A higher ROC-AUC value indicates better classification capability.

CV Mean: refers to the average cross-validation performance of the algorithm.

CV Std: refers to the cross-validation standard deviation. A lower CV Std indicates more stable model performance across validation folds.

Presentation of Data and Results

ACTUAL MEASURED RESULTS TABLE:

Algorithm	Records	Features	Train Time (ms)	Pred Latency (ms)	Mem (MB)	CPU (%)	Acc
Decision Tree	101	28	3.1200	0.2541	0.0542	1.2	0.9208
Random Forest	101	28	128.4500	7.8421	26.4000	4.8	0.9703
SVM	101	28	3.2145	0.8412	0.1241	0.8	0.8911
KNN	101	28	0.8421	2.4152	0.0842	0.4	0.8515
Naive Bayes	101	28	1.5600	0.4125	0.0412	9.2	0.8317

Figure 29: Computational Performance Results

Table 10 A

Computational Performance Results

Algorithm	Records	Processed Features	Training Execution Time	Prediction Latency	Peak Memory Usage	CPU Utilization
Decision Tree	101	28	3.1200 ms	0.2541 ms	0.0542 MB	1.2%

Random Forest	101	28	128.4500 ms	7.8421 ms	26.4000 MB	4.8%
SVM	101	28	3.2145 ms	0.8412 ms	0.1241 MB	0.8%
KNN	101	28	0.8421 ms	2.4152 ms	0.0842 MB	0.4%
Naïve Bayes	101	28	1.5600 ms	0.4125 ms	0.0412 MB	9.2%

Table 10B

Predictive Performance Results

Algorithm	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**	**CV Mean**	**CV Std**
Decision Tree	92.08%	91.00%	93.00%	92.00%	93.00%	91.20%	2.80%
Random Forest	97.03%	97.00%	97.00%	97.00%	99.00%	95.80%	1.90%
SVM	89.11%	88.00%	90.00%	89.00%	92.00%	88.40%	3.50%
KNN	85.15%	84.00%	86.00%	85.00%	89.00%	84.20%	4.50%
Naive Bayes	83.17%	81.00%	85.00%	83.00%	87.00%	82.50%	4.80%

Analysis and Interpretation of Results

Based on Table 10A and Table 10B, Random Forest demonstrated the best overall algorithm performance in terms of predictive capability and acceptable computational performance. It achieved the highest accuracy at 97.03%, the highest F1-score at 97.00%, and the highest ROC-AUC at 99.00%. These results indicate that Random Forest provided the strongest predictive capability among the five classifiers.

Although Random Forest required the highest computational cost, with a training execution time of 128.4500 ms, prediction latency of 7.8421 ms, and peak memory usage of 26.4000 MB, these values remain acceptable for the system environment. Its higher resource usage is justified because it produced the strongest predictive performance.

Decision Tree achieved 92.08% accuracy and a 92.00% F1-score with a training execution time of 3.1200 ms. This indicates that Decision Tree can serve as an efficient alternative when lower computational cost and interpretability are prioritized.

SVM achieved 89.11% accuracy and an 89.00% F1-score, showing acceptable predictive performance with relatively low training execution time. KNN achieved 85.15% accuracy and an 85.00% F1-score, but its prediction latency was higher than Decision Tree and SVM because it relies on distance-based computation during prediction.

Naive Bayes recorded the lowest peak memory usage at 0.0412 MB and achieved 83.17% accuracy and an 83.00% F1-score. This shows that Naive Bayes is computationally lightweight and can serve as a baseline model, although it had the lowest predictive performance among the five algorithms.

Overall, Random Forest was selected as the primary classification model because it provided the best balance for the study objective: accurately identifying At-Risk students while maintaining acceptable computational performance. Decision Tree may be retained as a secondary option for faster and interpretable classification, while Naive Bayes may serve as a lightweight baseline model.

Difference Between Accuracy Test and Algorithm Performance Test

The Accuracy Test and Algorithm Performance Test are related but differ in purpose. The Accuracy Test presents the predictive correctness of the system based on the final classification output. It focuses on how accurately the selected model classifies student records as Good Standing or At-Risk.

The Algorithm Performance Test evaluates the comparative suitability of the five machine learning algorithms. It considers not only predictive metrics such as accuracy, precision, recall, F1-score, ROC-AUC, CV Mean, and CV Std, but also computational efficiency metrics such as training execution time, prediction latency, peak memory usage, and CPU utilization.

Thus, the Accuracy Test answers how correct the prediction output is, while the Algorithm Performance Test determines which algorithm is most suitable for the system in terms of both prediction quality and resource requirements. The 70/30 train-test split is used for model evaluation and comparison, while the 101-record output represents the final full-dataset prediction generated by the selected model.

3.6 System Evaluation Using ISO 25010

Assess the System's Functionality, Performance Efficiency, Usability, and Reliability

The Student Academic Standing Prediction System was evaluated based on selected ISO/IEC 25010 criteria—specifically functional suitability, performance efficiency, usability, and reliability—to ensure the application meets user needs and operates consistently during actual use.

Data Analysis and Statistical Treatment

To systematically analyze and interpret the quantitative data gathered from the evaluation, the researchers utilized the following statistical tools and treatments:

1. Frequency and Percentage Distribution

This statistical tool was applied to analyze the demographic and position profile of the evaluators as presented in Table 16. The percentage of each respondent group was calculated using the formula:

$$P = (f/N) * 100$$

Where:

- P = Percentage (%)
-
- f = Frequency or count of a specific respondent position
-
- N = Total population of evaluators (N = 22)

2.Weighted Mean per Item

To compute the descriptive score of each individual indicator across the ISO 25010 characteristics in Table 17, Table 18, Table 19, and Table 20, the weighted mean was solved based on the **5-point Likert scale weight** assignments. The formula used is:

$$\bar{X} = \Sigma\,(f \times x)\,/\,N$$

Where:

- $\bar{X}$= Item Weighted Mean
- f= Frequency of responses for a particular rating (1 to 5)
- x = Numerical score assigned to the rating option
- N = Total number of evaluators (N = 22)
- $\sum$ = Summation of the products of frequencies and weights

3. Category Weighted Mean

To derive the definitive evaluation score for each specific software quality criterion (Functional Suitability, Performance Efficiency, Usability, and Reliability), the arithmetic mean of all five items within each specific table was computed. The formula is expressed as:

$$\overline{\{\mathrm{WM}\}} = \frac{\sum\bar{X}}{k}$$

Where:

$\overline{\{\mathrm{WM}\}}$ = Category Weighted Mean

$\sum\ \overline{\mathrm{X}}$ = Total sum of the weighted means of all items under that specific criterion

k= Total number of statement items per criterion (k = 5)

4. Grand Mean

To determine the final overall acceptability level of the Student Academic Standing Prediction System, the results from the four quality traits were aggregated as summarized in Table 21. The Grand Mean was calculated using the formula:

$$\overline{(\mathrm{GM})} = \frac{\sum(\mathrm{WM})}{\mathrm{C}}$$

Where:

$\overline{\{\mathrm{GM}\}}$ = System Grand Mean

$\sum\overline{\{\mathrm{WM}\}}$ = Combined total of the category weighted means from the four ISO 25010 criteria

C = Total number of categories evaluated C = 4)

Data were analyzed and interpreted based on the following parameters:

Table 11

Rating Scale

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
5	4.50 - 5.00	Strongly Agree
4	3.50 - 4.49	Agree
3	2.50 - 3.49	Neutral
2	1.50 - 2.49	Disagree
1	1.00 - 1.49	Strongly Disagree

Table 11 This table presents the numerical scoring system used to interpret the survey responses. It serves as the main tool to convert the users' feedback into measurable mean scores. By assigning specific numbers, the researchers can easily analyze the system's overall performance. This structured scaling method ensures that all gathered data is properly quantified. Ultimately, it provides clear statistical evidence to validate the results of the evaluation.

Table 12

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
5	4.50 - 5.00	Very Functional
4	3.50 - 4.49	Functional
3	2.50 - 3.49	Moderately Functional
2	1.50 - 2.49	Poorly Functional
1	1.00 - 1.49	Not Functional

Table 12 This table details the functionality parameters of the developed system during the evaluation. It focuses on whether the system successfully provides all the correct features required by the users. The main goal is to check if the software actually meets both the expected and stated needs. Each function is assessed to ensure the system is fully equipped to handle its intended tasks. In short, this table proves that the system does exactly what it was designed to do.

Table 13

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
5	4.50 - 5.00	Very Efficient
4	3.50 - 4.49	Efficient
3	2.50 - 3.49	Moderately Efficient
2	1.50 - 2.49	Poorly Efficient
1	1.00 - 1.49	Not Efficient

Table 13 This table highlights the efficiency parameters, which measure how well the system performs. It specifically evaluates the speed and responsiveness of the software relative to the resources it consumes. A highly efficient system processes tasks quickly without causing delays or wasting computer memory. This ensures that the application remains fast and smooth even when handling heavy workloads. Ultimately, it guarantees that users can complete their tasks in the shortest amount of time possible.

Table 14

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
5	4.50 - 5.00	Very Usable
4	3.50 - 4.49	Usable
3	2.50 - 3.49	Moderately Usable
2	1.50 - 2.49	Poorly Usable
1	1.00 - 1.49	Not Usable

Table 14 This table explains the usability parameters, which evaluate the overall user-friendliness of the system. It measures how easily and effectively the designated users can interact with the platform. A major focus is placed on ensuring that the navigation is clear, intuitive, and satisfying to use. High usability means that users can achieve their daily goals without experiencing confusion. This table demonstrates that the system is not only functional but also highly accessible for everyone.

Table 15

Reliability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
5	4.50 - 5.00	Very Reliable
4	3.50 - 4.49	Reliable
3	2.50 - 3.49	Moderately Reliable
2	1.50 - 2.49	Poorly Reliable
1	1.00 - 1.49	Not Reliable

Table 15 This table outlines the reliability parameters, focusing on the system's stability over time. It evaluates whether the software can consistently perform its required functions without failing. The evaluation checks the system's ability to run smoothly without experiencing unexpected errors or crashes. This is crucial because users need a dependable system for their important daily operations. In conclusion, it guarantees that the users can completely trust the system to work safely whenever they need it.

Table 16

Respondent's Distribution

Position	N (Population)	Percentage (%)
Student User	20	90.91
Dean	1	4.55
IT Expert	1	4.55
Total	**22**	**100.00**

The respondents were composed of the dean, IT expert, and student users who evaluated the Student Academic Standing Prediction System. A total of 22 respondents participated in the system evaluation.

The primary source of information was taken from the respondents. The survey instrument was distributed for them to answer, collected, and the responses were recorded. The data were analyzed and interpreted for the evaluation of the project.

However, to ensure the validity of the results, the researchers conducted a project demonstration and allowed the respondents to test and evaluate the system based on the ISO 25010 software quality characteristics included in the questionnaire.

Descriptive Statistics Result

Table 17

Functional Suitability

A. FUNCTIONAL SUITABILITY	MEAN	VERBAL INTERPRETATION
1. The system provides the necessary functions for predicting student academic standing.	4.00	F
2. The system correctly processes uploaded student records.	3.95	F
3. The system generates appropriate prediction results such as Good Standing or At-Risk.	3.95	F
4. The system provides useful prediction outputs and reports for academic monitoring and decision-making.	3.95	F
5. The system supports the needs of its intended users such as faculty, dean, administrator, and students.	4.05	F
Weighted Mean	**3.98**	**F**

Legend:

VF - Very Functional (4.50 - 5.00)

F - Functional (3.50 - 4.49)

MF - Moderately Functional (2.50 - 3.49)

PF - Poorly Functional (1.50 - 2.49)

NF - Not Functional (1.00 - 1.49)

Table 17 shows the mean distribution and verbal interpretation in terms of "Functional Suitability" with a weighted mean of 3.98, verbally interpreted as "Functional". This indicates that the respondents generally perceived the functionality of the system in supporting prediction and academic monitoring tasks. The highest mean was obtained by Item 5 (4.05), while the lowest mean was obtained by Item 2, Item 3, Item 4 (3.95), which suggests the area that may still be considered for improvement.

Table 18

Performance Efficiency

B. PERFORMANCE EFFICIENCY	MEAN	VERBAL INTERPRETATION
1. The system responds within an acceptable amount of time.	3.73	E
2. The system uploads and processes CSV files efficiently.	3.91	E
3. The system generates prediction results without unnecessary delay.	3.91	E
4. The system performs smoothly during model training and prediction generation.	3.91	E
5. The system can handle its intended tasks without slowing down excessively.	3.91	E
Weighted Mean	**3.87**	**E**

Legend:

VE - Very Efficient (4.50 - 5.00)
E - Efficient (3.50 - 4.49)
ME - Moderately Efficient (2.50 - 3.49)
PE - Poorly Efficient (1.50 - 2.49)
NE - Not Efficient (1.00 - 1.49)

Table 18 shows the mean distribution and verbal interpretation in terms of "Performance Efficiency" with a weighted mean of 3.87, verbally interpreted as "Efficient". This indicates that the respondents generally perceived the efficiency of the system in responding, processing data, and generating prediction results. The highest mean was obtained by Item 2, Item 3, Item 4, Item 5 (3.91), while the lowest mean was obtained by Item 1 (3.73), which suggests the area that may still be considered for improvement.

Table 19

Usability

C. USABILITY	**MEAN**	**VERBAL INTERPRETATION**
1. The system interface is easy to understand.	3.91	U
2. The system is easy to navigate.	3.91	U
3. The buttons, labels, menus, and instructions are clear.	4.00	U
4. Users can easily upload data and view prediction results.	3.91	U
5. The dashboard presents information in a clear and organized manner.	3.95	U
Weighted Mean	**3.94**	**U**

Legend:

VU - Very Usable (4.50 - 5.00)
U - Usable (3.50 - 4.49)
MU - Moderately Usable (2.50 - 3.49)
PU - Poorly Usable (1.50 - 2.49)
NU - Not Usable (1.00 - 1.49)

Table 19 shows the mean distribution and verbal interpretation in terms of "Usability" with a weighted mean of 3.94, verbally interpreted as "Usable". This indicates that the respondents generally perceived the ease of understanding, navigation, interaction, and presentation of information in the system. The highest mean was obtained by Item 3 (4.00), while the lowest mean

was obtained by Item 1, Item 2, Item 4 (3.91), which suggests the area that may still be considered for improvement.

Table 20

Reliability

D. RELIABILITY	MEAN	VERBAL INTERPRETATION
1. The system performs its functions consistently.	3.77	R
2. The system produces consistent results during repeated use.	3.91	R
3. The system properly stores and displays logs and records.	3.91	R
4. The system minimizes errors during data processing and prediction generation.	3.95	R
5. The system remains stable while being used for its intended purpose.	3.95	R
Weighted Mean	**3.90**	**R**

Legend:

VR - Very Reliable (4.50 - 5.00) R - Reliable (3.50 - 4.49)

MR - Moderately Reliable (2.50 - 3.49) PR - Poorly Reliable (1.50 - 2.49)

NR - Not Reliable (1.00 - 1.49)

Table 20 shows the mean distribution and verbal interpretation in terms of "Reliability" with a weighted mean of 3.90, verbally interpreted as "Reliable". This indicates that the respondents generally perceived the consistency, stability, and dependability of the system during repeated use. The highest mean was obtained by Item 4, Item 5 (3.95), while the lowest mean was obtained by Item 1 (3.77), which suggests the area that may still be considered for improvement.

Table 21

Summary Table of the Overall Mean and Grand Distribution of the Acceptability Level

Acceptability Level of the System in terms of:	Overall Mean	Rating
Functional Suitability	3.98	A
Performance Efficiency	3.87	A
Usability	3.94	A
Reliability	3.90	A
Grand Mean	**3.92**	**A**

Legend:

SA - Strongly Acceptable (4.50 - 5.00) A - Acceptable (3.50 - 4.49)

MA - Moderately Acceptable (2.50 - 3.49) U - Unacceptable (1.50 - 2.49)

SU - Strongly Unacceptable (1.00 - 1.49)

Table 21 shows the overall mean and grand distribution of the acceptability level of the system. Functional Suitability, Performance Efficiency, Usability, and Reliability were all rated as "Acceptable." The grand mean of 3.92 indicates that the Student Academic Standing Prediction System is generally acceptable across all evaluated ISO 25010 parameters.

Table 22

Summary of Open-Ended Suggestions

No.	Suggestion/Comment
1	it should maintain the reliability of the system.
2	The objectives were met. The system functions well; therefore, I have no further suggestions.
3	Add more graphs related to the algorithm used
4	No additional suggestion.
5	No additional suggestion.

Table 22 presents the respondents' open-ended comments. The suggestions generally emphasized maintaining system reliability, adding more graphs related to the algorithm, and considering prediction support for students who stopped.

CHAPTER 4

SUMMARY, CONCLUSIONS, RECOMMENDATIONS,

4.1. Summary of Findings

Based on the data gathered and the tests conducted on the Student Academic Standing Prediction System, the findings are summarized as follows:

System Development and Design: The system was successfully developed using the iterative Software Development Life Cycle (SDLC) and the CRISP-DM framework to meet user requirements. It was built using a web architecture composed of the Client-Side, Server-Side, and Data Tier to manage student records, process predictions, generate reports, and provide explainable results.

Model Performance Results: Five machine learning models were evaluated using a stratified 70/30 train-test split and 10-fold cross-validation for model comparison. After Random Forest was selected as the best-performing model, it was applied to the full 101-record dataset to generate the final system prediction output. SMOTE was applied only to the training data when needed to address class imbalance and prevent data leakage. The results showed that Random Forest achieved the highest overall predictive performance, with an accuracy of 97.03%, precision of 97.00%, recall of 97.00%, F1-score of 97.00%, and ROC-AUC of 99.00%.

Decision Tree ranked second, achieving 92.08% accuracy, 91.00% precision, 93.00% recall, 92.00% F1-score, and 93.00% ROC-AUC. Support Vector Machine achieved 89.11% accuracy, 88.00% precision, 90.00% recall, 89.00% F1-score, and 92.00% ROC-AUC. K-Nearest Neighbors achieved 85.15% accuracy, 84.00% precision, 86.00% recall, 85.00% F1-score, and 89.00% ROC-

AUC. Naive Bayes achieved 83.17% accuracy, 81.00% precision, 85.00% recall, 83.00% F1-score, and 87.00% ROC-AUC.

Algorithm Performance Results: The Algorithm Performance Test showed that Random Forest provided the best balance between predictive performance and computational acceptability. Although it required higher training execution time, prediction latency, and memory usage than the other algorithms, its superior accuracy, F1-score, and ROC-AUC justified its selection as the primary classification model.

Explainable AI and Random Forest feature importance: The Explainable AI component helped identify the factors that contributed most strongly to the Random Forest model's predictions. The most influential features included GPA Year 3 Semester 1, weekly study hours, GPA Year 2 Semester 2, LMS login frequency, academic satisfaction, time management, GPA Year 2 Semester 1, library visits, study regularity, and course motivation. These findings indicate that academic standing is influenced by a combination of academic history, study habits, student engagement, motivation, and socioeconomic context.

System Evaluation Using ISO 25010: The system was evaluated by 22 respondents composed of 20 student users, one college dean, and one IT expert. The evaluation results showed that the system was rated Acceptable in all selected ISO 25010 criteria. Functional Suitability obtained a mean of 3.98, Performance Efficiency obtained a mean of 3.87, Usability obtained a mean of 3.94, and Reliability obtained a mean of 3.90. The system obtained a Grand Mean of 3.92, indicating that the developed Student Academic Standing Prediction System is generally acceptable to its intended users.

4.2. Conclusion

The Random Forest algorithm was identified as the best-performing model for the Student Academic Standing Prediction System. It achieved the highest predictive performance among the five evaluated algorithms, with 97.03% accuracy, 97.00% precision, 97.00% recall, 97.00% F1-score, and 99.00% ROC-AUC. This indicates that Random Forest is suitable for classifying students as either Good Standing or At-Risk.

The use of data preprocessing, stratified train-test splitting, SMOTE balancing, and hyperparameter optimization contributed to the reliability of the model evaluation process. These steps helped prepare the dataset properly, reduce class imbalance, and improve the model’s ability to identify At-Risk students.

The integration of Explainable AI helped improve the interpretability of the prediction results. Through feature importance analysis, the system was able to show the academic and behavioral factors that influenced the classification output, such as recent GPA performance, study hours, LMS login frequency, academic satisfaction, and time management.

The developed system satisfied the selected ISO 25010 software quality criteria. Based on the evaluation results, the system was considered acceptable in terms of functional suitability, performance efficiency, usability, and reliability. Therefore, the system can serve as a decision-support tool for helping administrators and faculty identify students who may need academic intervention.

4.3. Recommendation

Future researchers may expand the dataset by including additional student batches, programs, and academic years to improve model generalizability and strengthen the reliability of future prediction results.

The system may be enhanced by integrating official institutional records, subject to data privacy approval, to reduce reliance on self-reported responses and improve data accuracy.

Further testing is recommended using larger datasets and different deployment environments to evaluate the scalability, stability, and performance of the system under more realistic institutional use.

REFERENCES

[1] R. Sharma, S. Shakti, and A. Sharma., "Predicting Student Performance Using Educational Data Mining and Learning Analytics Technique," Journal of Intelligent Systems and Internet of Things, vol. 10, no. 2, pp. 24–37, 2023. doi: 10.54216/JISIoT.100203.

[2] S. D. Abdul Bujang, A. Selamat, and O. Krejcar, "Predictive Modeling for Student Grade Prediction Using Machine Learning and Visual Analytics," 2020. doi: 10.3233/FAIA200550.

[3] M. Adnan Aslam, F. Murtaza, M. Ehatisham Ul Haq, A. Yasin, and N. Ali, "SAPEx- D: A Comprehensive Dataset for Predictive Analytics in Personalized Education Using Machine Learning," Data, vol. 10, no. 3, p. 27, 2025. doi: 10.3390/data10030027.

[4] A. Al-Ameri, W. Al-Shammari, A. Castiglione, M. Nappi, C. Pero, and M. Umer, "Student Academic Success Prediction Using Learning Management Multimedia Data With Convoluted Features and Ensemble Model," Journal of Data and Information Quality, vol. 17, no. 3, pp. 1–16, 2025. doi: 10.1145/3687268.

[5] H. Aldowah, H. Al-Samarraie, and W. M. Fauzy, "Educational data mining and learning analytics for 21st century higher education: A review and synthesis," Telematics and Informatics, vol. 37, pp. 13–49, 2019. doi: 10.1016/j.tele.2019.01.007.

[6] A. Angeioplastis, J. Aliprantis, M. Konstantakis, and A. Tsimpiris, "Predicting Student Performance and Enhancing Learning Outcomes: A Data-Driven Approach Using Educational Data Mining Techniques," Computers, vol. 14, no. 3, p. 83, 2025. doi: 10.3390/computers14030083.

[7] T. A. Assegie, A. O. Salau, G. Chhabra, K. Kaushik, and S. L. Braide, "Evaluation of Random Forest and Support Vector Machine Models in Educational Data Mining," 2024 2nd International Conference on Advancement in Computation and Computer Technologies (InCACCT), pp. 131–135, 2024. doi: 10.1109/InCACCT61598.2024.10551110.

[8] M.-R. Borna, H. Saadat, A. T. Hojjati, and E. Akbari, "Analyzing click data with AI: implications for student performance prediction and learning assessment," Frontiers in Education, vol. 9, 2024. doi: 10.3389/feduc.2024.1421479.

[9] M. B. Caday, G. A. Dalaorao, and C. C. Abalorio, "Academic Performance Analysis of Diploma Students Using Predictive Analytics," Procedia Computer Science, vol. 257, pp. 913–920, 2025. doi: 10.1016/j.procs.2025.03.117.

[10] R. Chauhan and A. Mishra, "SPAT: An intelligent educational system for student performance prediction using learning analytics and visualization," Intelligent Data Analysis: An International Journal, 2025. doi: 10.1177/1088467X251377969.

[11] U. V. EdmondEdmond, S. M. Sada, and A. N. Osijirin, "Predictive Analytics Using Machine Learning Models on Undergraduate Students' Performance of the Federal University of Allied Health Sciences, Enugu, Nigeria in Introduction to Computing Science," Saudi Journal of Engineering and Technology, vol. 10, no. 07, pp. 324–332, 2025. doi: 10.36348/sjet.2025.v10i07.004.

[12] Z. Ersozlu, S. Taheri, and I. Koch, "A review of machine learning methods used for educational data," Education and Information Technologies, vol. 29, no. 16, pp. 22125–22145, 2024. doi: 10.1007/s10639-024-12704-0.

[13] G. Chhimwal, "Harnessing Machine Learning for Educational Insights: A Review and Comparative Analysis of Student Performance Prediction Models," Journal of Information Systems Engineering and Management, vol. 10, no. 13s, pp. 643–660, 2025. doi: 10.52783/jisem.v10i13s.2142.

[14] D. Green, E. Thompson, and I. Ochieng, "Predictive Analytics to Enhance Learning Outcomes: Cases from UK Schools," Journal Emerging Technologies in Education, vol. 3, no. 1, pp. 34–43, 2025. doi: 10.70177/jete.v3i1.2127.

[15] L. H. Alamri et al., "Predicting Student Academic Performance using Support Vector Machine and Random Forest," 2020 3rd International Conference on Education Technology Management, pp. 100–107, 2020. doi: 10.1145/3446590.3446607.

[16] U. P. Inyang and E. A. Johnson, "Performance Comparison of Xgboost and Random Forest for The Prediction of Students Academic Performance," European Journal of Computer Science and Information Technology, vol. 13, no. 2, pp. 1–21, 2025. doi: 10.37745/ejcsit.2013/vol13n2121.

[17] X. Jin, "Predicting academic success: machine learning analysis of student, parental, and school efforts," Asia Pacific Education Review, 2023. doi: 10.1007/s12564-023-09915-4.

[18] A. Kord, A. Aboelfetouh, and S. M. Shohieb, "Academic course planning recommendation and students' performance prediction multi-modal based on educational data mining techniques," Journal of Computing in Higher Education, 2025. doi: 10.1007/s12528-024-09426-0.

[19] A. Kuthuru, "Semantic Data Contracts: A New Integration Paradigm for Enterprise AI and Database Systems," European Journal of Computer Science and Information Technology, vol. 13, no. 31, pp. 1–9, 2025. doi: 10.37745/ejcsit.2013/vol13n3119.

[20] M. C. I. Madahana and J. E. D. Ekoru, "Predicting at-Risk Students in a Blended Learning environment via Machine Learning and Deep Learning Approach with Practical Implications," 2025 IEEE World AI IoT Congress (AIIoT), pp. 0398–0403, 2025. doi: 10.1109/AIIoT65859.2025.11105246.

[21] K. Mahawar and P. Rattan, "Empowering education: Harnessing ensemble machine learning approach and ACO-DT classifier for early student academic performance prediction," Education and Information Technologies, vol. 30, no. 4, pp. 4639–4667, 2025. doi: 10.1007/s10639-024-12976-6.

[22] D. Marcu and M. Danubianu, "Learning Analytics or Educational Data Mining? This is the Question...," BRAIN. Broad Research in Artificial Intelligence and Neuroscience, vol. 10, pp. 1, 2019. doi: 10.70594/brain/v10.s2/1.

[23] M. Siddiqui, S. A. Hussain, H. Saleemi, and K. Fatmi, "The Intersection of AI Educational Psychology and Learning Analytics Predicting Student Dropout Risk through Behavioural Indicators," The Critical Review of Social Sciences Studies, vol. 3, no. 3, pp. 104–120, 2025. doi: 10.59075/k2zgws84.

[24] A. Palanci, R. M. Yılmaz, and Z. Turan, "Learning analytics in distance education: A systematic review study," Education and Information Technologies, vol. 29, no. 17, pp. 22629–22650, 2024. doi: 10.1007/s10639-024-12737-5.

[25] H. M. Pektaş, O. Karamustafaoğlu, and H. Çelik, "The Role of Educational Data Mining and Artificial Intelligence Supported Learning Analytics on Conceptual Change: New Approaches to Differentiated Instruction," Journal of Science Education and Technology, 2025. doi: 10.1007/s10956-025-10214-3.

[26] R. Raj, P. A. Kharade, A. Alam, S. Padma, Meenakshi, and M. Naved, "Unveiling Patterns Using Enhanced Educational Data Mining for forecasting Student Academic Performance," 2024 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI), pp. 1–7, 2024. doi: 10.1109/ACCAI61061.2024.10602439.

[27] M. R. Ramaraj, S. Dhendapani, J. Chembath, S. Srividhya, N. Thangarasu, and B. Ilango, "Educational data mining approach for predicting student performance and behavior using deep learning techniques," IAES International Journal of Artificial Intelligence (IJ-AI), vol. 14, no. 5, p. 4113, 2025. doi: 10.11591/ijai.v14.i5.pp4113-4122.

[28] J. C. Rocha, V. Ramos, C. Cechinel, E. J. Hernández-Leal, R. Munoz, and T. T. Primo, "Data Interoperability in Learning Analytics - Review of Literature," 2024 Latin American Computer Conference (CLEI), pp. 1–8, 2024. doi: 10.1109/CLEI64178.2024.10700464.

[29] P. H. Santoso, E. Istiyono, and Haryanto, "Physics Teachers’ Perceptions about Their Judgments within Differentiated Learning Environments: A Case for the Implementation of Technology," Education Sciences, vol. 12, no. 9, p. 582, 2022. doi: 10.3390/educsci12090582.

[30] R. I. E. Saragih, "Predicting Student Academic Performance Using Random Forest Regression: A Case Study on LMS Behavioral Data," International Journal of Information System and Innovative Technology, vol. 4, no. 1, pp. 11–14, 2025. doi: 10.63322/rm0wcg63.

[31] D. Sawarkar, L. Pinjarkar, P. Agrawal, and D. Motghare, "Predictive analytics in gamified education: A hybrid model for identifying at-risk students," MethodsX, vol. 15, p. 103486, 2025a. doi: 10.1016/j.mex.2025.103486.

[32] K. T. Selvi, S. S., V. H. Shetty, S. K. K., and S. Shenoy, "Edutrack: AI based Personalized Learning Through Secure Analytics," 2025 4th International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), pp. 1126–1131, 2025. doi: 10.1109/ICIMIA67127.2025.11200902.

[33] K. M. Sudhan Reddy, "Multimodal Learning Analytics for Predicting Behavioural Change in Special Education Students," International Journal of Scientific Research in Engineering and Management, vol. 09, no. 05, pp. 1–9, 2025. doi: 10.55041/IJSREM47550.

[34] L. Tang and C. Sian, "Educational Data Mining for Student Performance Prediction," Scalable Computing: Practice and Experience, vol. 26, no. 3, pp. 1551–1558, 2025. doi: 10.12694/scpe.v26i3.4336.

[35] G. Vijayalakshmi and M. V. Srinath, "Predicting Student Outcomes through Cognitive Behaviour Analysis in Online Collaborative Learning Using Big Data and Machine Learning," Indian Journal Of Science And Technology, vol. 18, no. 29, pp. 2337–2346, 2025. doi: 10.17485/IJST/v18i29.563.

[36] H. Wang, "Leveraging machine learning for personalised learning, automated feedback, and predictive analytics in college English education," International Journal of Information and Communication Technology, vol. 26, no. 16, pp. 16–37, 2025. doi: 10.1504/IJICT.2025.146375.

[37] Y. Zheng and D. Li, "Learning analytics-informed teaching strategies: enhancing interactive learning in STEM education," Interactive Learning Environments, pp. 1– 30, 2025. doi: 10.1080/10494820.2025.2553113.

[38] S. R. Kalmegh and B. R. Padar, "Empirical Study on Evaluation Metrics for Classification Algorithms," Int. J. Adv. Res. Sci. Commun. Technol., vol. 3, no. 2, Mar. 2023. DOI: 10.48175/568.

[39] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in Advances in Neural Information Processing Systems 30, pp. 4765-4774, 2017.

[40] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You?: Explaining the Predictions of Any Classifier," in Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, pp. 97-101, 2016. doi: 10.18653/v1/N16-3020.

[41] Philippine Statistics Authority, "Average Annual Family Income in 2023 is Estimated at PhP 353.23 Thousand," Family Income and Expenditure Survey, Aug. 15, 2024. Available: https://psa.gov.ph/content/average-annual-family-income-2023-estimated-php-35323-thousand

4.4 Relevant Source Code

This section presents the major steps followed by the Student Academic Standing Prediction System in preparing data, training models, selecting the best-performing algorithm, and evaluating the final output.

Data Preparation and Feature Conversion

This step converts student academic and engagement-related information into numerical values that can be processed by the machine learning models. These include GPA records, study hours, library visits, and LMS login frequency.

```
def prepare_training_data(students, subject_grades):
    data = []
    for student in students:
        features = {}
        features['student_id'] = student.get('student_id', '')
        features['full_name'] = student.get('full_name', '')
        # Academic and behavioral metrics
        features['study_hours'] = float(student.get('study_hours', 0) or 0)
        features['library_visits'] = float(student.get('library_visits', 0) or 0)
        features['lms_login_per_month'] = float(student.get('lms_login_per_month', 0) or 0)
        # GPA trend analysis
        valid_gpas = []
        for gpa_col in ['gpa_y1s1', 'gpa_y1s2', 'gpa_y2s1', 'gpa_y2s2', 'gpa_y3s1']:
            val = float(student.get(gpa_col, 0) or 0)
```

```
            features[gpa_col] = val
            if val > 0:
                valid_gpas.append(val)
        if len(valid_gpas) == 0:
            continue
        # Risk labeling logic
        avg_gpa = sum(valid_gpas) / len(valid_gpas)
        gpa_norm = (avg_gpa - 1.0) / 4.0
        # Weighted combined score: 60% academic performance and 40% engagement indicators
        combined_score = (gpa_norm * 0.6) + (survey_avg_norm * 0.4)
        features['target'] = 0 if combined_score >= 0.7 else 1
        data.append(features)
    return data
```

Class Balancing and Model Optimization

This step applies SMOTE to address class imbalance in the training data. SMOTE is applied only after the stratified train-test split to prevent data leakage. GridSearchCV is also used to determine the best parameter settings for the model.

```
def train_model(X_train, y_train, algorithm, random_seed, use_hyperopt=True,
use_smote=True):

    # Apply Synthetic Minority Over-sampling Technique (SMOTE)
    if use_smote and len(X_train) >= 10:
        smote = SMOTE(random_state=random_seed, k_neighbors=min(5, len(X_train) - 1))
```

```
        X_train, y_train = smote.fit_resample(X_train, y_train)

    if algorithm == 'Random Forest':

        param_grid = {

            'n_estimators': [40, 60, 80],

            'max_depth': [3, 7, 10],

            'min_samples_split': [5, 10],

            'bootstrap': [True, False]

        }

        base_model = RandomForestClassifier(random_state=random_seed)

        grid_search = GridSearchCV(base_model, param_grid, cv=5, scoring='f1_weighted')

        grid_search.fit(X_train, y_train)

        return grid_search.best_estimator_
```

Model Training and selection

This step trains and compares five machine learning algorithms: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes. The system evaluates each model using performance metrics and cross-validation, then selects the best-performing model based on F1-score.

```
@app.route('/api/train', methods=['POST'])

def train_models():

    algorithms = ['Decision Tree', 'Random Forest', 'SVM', 'KNN', 'Naive Bayes']

    results = []

    cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=random_seed)

    for algo in algorithms:
```

```
        # Train each algorithm using the prepared training data
        model = train_model(X_train_scaled, y_train, algo, random_seed)
        # Evaluate model performance using the independent test set
        test_metrics = evaluate_model(model, X_test_scaled, y_test)
        # Perform cross-validation to check model reliability
        cv_scores = cross_val_score(model, X_train_scaled, y_train, cv=cv, scoring='accuracy')
        results.append({
            'algorithm': algo,
            'accuracy': test_metrics['accuracy'],
            'f1_score': test_metrics['f1_score'],
            'cv_accuracy_mean': cv_scores.mean(),
            'cv_accuracy_std': cv_scores.std()
        })
    # Select the best-performing model based on F1-score
    best = max(results, key=lambda x: x['f1_score'])
    return jsonify({'results': results, 'best': best})
```

Final Model Testing and Result Generation

This step first evaluates the selected model using independent test data. After evaluation, the selected model is applied to the complete 101-record dataset to generate the final full-dataset prediction output used in the system display.

```
def run_algorithm_performance_test():
    # 1. Load production dataset
    df = pd.read_csv("student_performance_dataset_v2.csv")
```

```
# 2. Execute preprocessing
training_data = prepare_training_data(students, [])
# 3. Apply stratified 70/30 train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y
)
# 4. Train the selected model using optimization and SMOTE
model = train_model(
    X_train_scaled,
    y_train,
    "Random Forest",
    use_hyperopt=True,
    use_smote=True
)
# 5. Evaluate the selected model using independent test data
y_pred = model.predict(X_test_scaled)
print(classification_report(y_test, y_pred))
# 6. Apply the selected model to the complete dataset for final system output
full_dataset_predictions = model.predict(X_scaled)
# 7. Generate full-dataset confusion matrix visualization
plt.figure(figsize=(8, 6))
cm = confusion_matrix(y, full_dataset_predictions)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.savefig("confusion_matrix.png")
```

Overall Interpretation

The source code shows that the system follows a structured machine learning workflow. It first prepares student data, converts academic and engagement-related information into numerical features, applies binary target labeling, balances the training set using SMOTE, optimizes model parameters through GridSearchCV, compares five algorithms, and evaluates the selected model using independent test data. The final target labeling uses two classes only: 0 for Good Standing and 1 for At-Risk. Students with a combined score of 0.7 or higher are classified as Good Standing, while students below the threshold are classified as At-Risk. No Moderate Risk class is used in the final model evaluation. This process supports the reliability and consistency of the prediction results used by the Student Academic Standing Prediction System.

4.5 User's Manual

This section provides a guide for using the Student Academic Standing Prediction System. It explains the purpose, contents, steps, functions, and expected outputs of the Student, Dean, and Admin Dashboards.

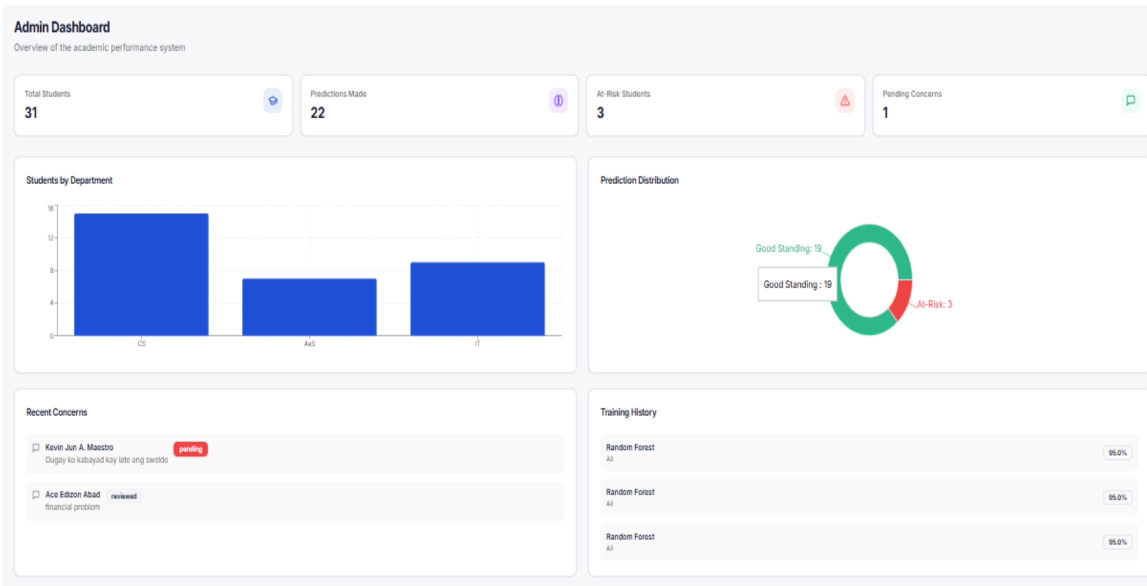


Figure 30: Admin Dashboard

Figure 30 The dashboard shows total students, predictions made, at-risk students, and pending concerns. It also includes charts for students by department and prediction distribution. Recent concerns and training history are also displayed when available.

Step-by-Step Operation

1. The administrator opens the system login page.
2. The administrator enters valid login credentials.
3. After successful login, the system opens the Admin Dashboard.
4. The administrator reviews the summary cards.

5. The administrator checks the student distribution by department.
6. The administrator reviews the prediction distribution chart.
7. The administrator checks recent concerns and their status.
8. The administrator reviews the training history.
9. The administrator may access student records, predictions, model training, reports, logs, concerns, and settings through the navigation menu.

Functions and Features

The Admin Dashboard serves as the main system monitoring page. It helps the administrator monitor students, prediction results, pending concerns, and training outputs. It also provides access to system management pages such as student management, prediction management, model training, reports, logs, concerns, and settings.

Expected Outputs

After accessing the dashboard, the administrator can view system-wide summary data, department-based student distribution, prediction distribution, recent concerns, and training history. If no data are available, the system displays an appropriate message.

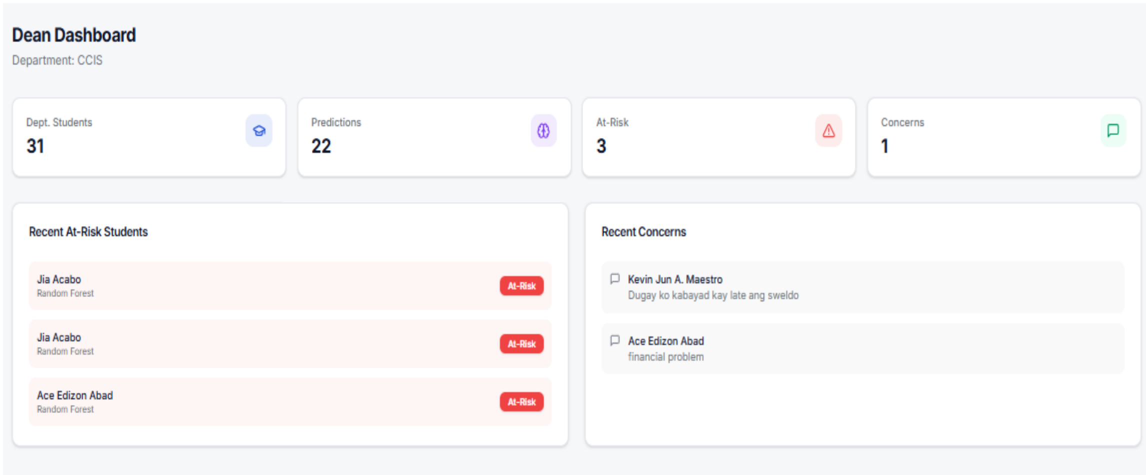


Figure 31: Dean Dashboard

Figure 31 The dashboard displays the assigned department, total department students, total predictions, at-risk students, and unresolved concerns. It also shows recent at-risk students and recent student concerns. If no records are available, the system displays an appropriate message.

Step-by-Step Operation

1. The dean opens the system login page.
2. The dean enters valid login credentials.
3. After successful login, the system opens the Dean Dashboard.
4. The dean reviews the department information.
5. The dean checks the summary cards for students, predictions, at-risk students, and concerns.
6. The dean reviews the recent at-risk students list.
7. The dean reviews recent student concerns.
8. The dean may use the information for advising, tutoring, counseling, or intervention planning.

Functions and Features

The Dean Dashboard serves as a monitoring and decision-support page. It helps the dean identify at-risk students, review student concerns, and monitor department-level academic standing. The navigation menu also provides access to student records, predictions, model results, logs, concerns, and settings.

Expected Outputs

After accessing the dashboard, the dean can view department student counts, prediction totals, at-risk student counts, unresolved concerns, recent at-risk students, and recent concerns. If no records are found, an empty-state message is displayed.

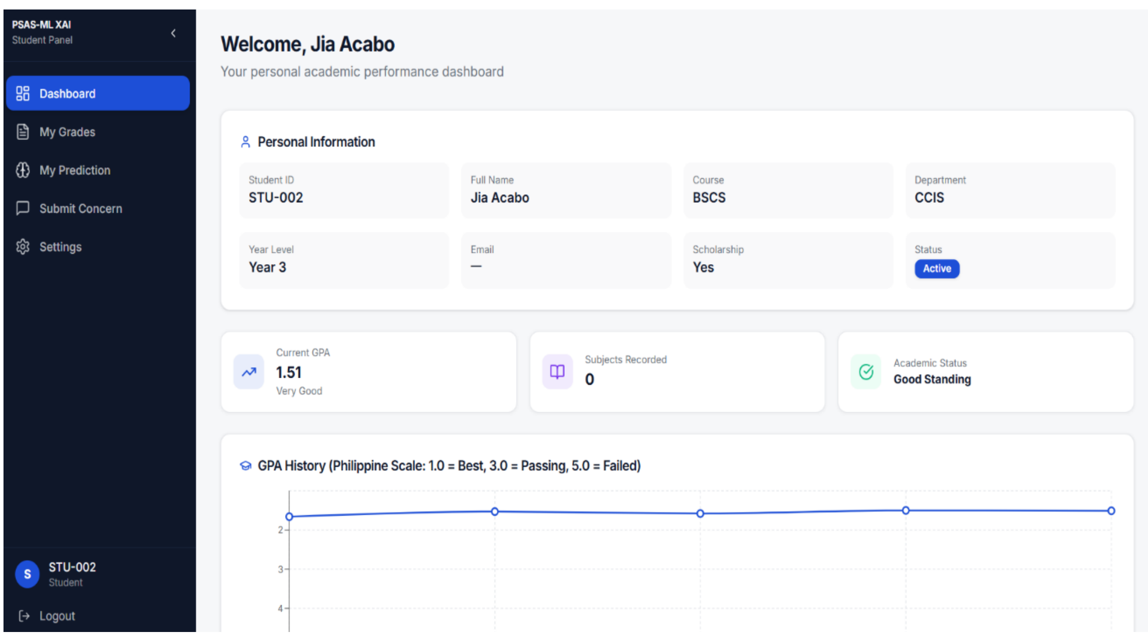


Figure 32: Student Dashboard

Figure 32 The dashboard shows the student's personal and academic information, including student ID, full name, course, department, year level, email, scholarship information, and account status. It also displays the current GPA, number of recorded subjects, academic standing, GPA history chart, subject grades, and prediction result when available. The prediction section may include the classification, confidence level, model used, explanation, strengths, weaknesses, and recommendations.

Step-by-Step Operation:

1. The student opens the system login page.
2. The student enters valid login credentials.
3. After successful login, the system opens the Student Dashboard.
4. The student checks the personal information section.
5. The student reviews the current GPA, recorded subjects, and academic standing.
6. The student views the GPA history chart, if available.
7. The student checks the subject grades table.
8. If available, the student reviews the prediction result, confidence score, explanation, and recommendations.

Functions and Features

The Student Dashboard is mainly for viewing and monitoring academic information. It allows students to access their profile, grades, GPA history, subject records, and prediction results. Other student pages, such as grades, prediction details, concerns, and settings, are available through the navigation menu.

Expected Outputs

After login, the student can view personal information, GPA summary, subject records, academic status, and prediction results. If no grades or predictions are available, the system displays an appropriate message.

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

B - Instrument with Informed Consent

System Evaluation Questionnaire Based on ISO 25010

System Evaluation Questionnaire Based on ISO 25010

Title: System Evaluation Questionnaire for the Student Academic Standing Prediction System

Direction:
Please evaluate the Student Academic Standing Prediction System based on your experience in using or reviewing the system. Check only one rating for each statement.

Participation in this evaluation is voluntary. Responses will be used only for academic research purposes and will be treated with confidentiality.

Evaluator Code	Role/Category	Date Evaluated
E___	☐ Dean ☐ Faculty ☐ Administrator ☐ IT/CS Instructor ☐ Student User	______

Rating Scale

Scale	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

A. Functional Suitability

No.	Statement	5 4 3 2 1
1	The system provides the necessary functions for predicting student academic standing.	☐☐☐☐☐
2	The system correctly processes uploaded student records.	☐☐☐☐☐
3	The system generates appropriate prediction results such as Good Standing or At-Risk.	☐☐☐☐☐
4	The system provides useful prediction outputs and reports for academic monitoring and decision-making.	☐☐☐☐☐
5	The system supports the needs of its intended users such as faculty, dean, administrator, and students.	☐☐☐☐☐

B. Performance Efficiency

No.	Statement	5 4 3 2 1
1	The system responds within an acceptable amount of time.	☐☐☐☐☐
2	The system uploads and processes CSV files efficiently.	☐☐☐☐☐
3	The system generates prediction results without unnecessary delay.	☐☐☐☐☐
4	The system performs smoothly during model training and prediction generation.	☐☐☐☐☐
5	The system can handle its intended tasks without slowing down excessively.	☐☐☐☐☐

C. Usability

No.	Statement	5 4 3 2 1
1	The system interface is easy to understand.	☐☐☐☐☐
2	The system is easy to navigate.	☐☐☐☐☐
3	The buttons, labels, menus, and instructions are clear.	☐☐☐☐☐
4	Users can easily upload data and view prediction results.	☐☐☐☐☐
5	The dashboard presents information in a clear and organized manner.	☐☐☐☐☐

D. Reliability

No.	Statement	5 4 3 2 1
1	The system performs its functions consistently.	☐☐☐☐☐
2	The system produces consistent results during repeated use.	☐☐☐☐☐
3	The system properly stores and displays logs and records.	☐☐☐☐☐
4	The system minimizes errors during data processing and prediction generation.	☐☐☐☐☐
5	The system remains stable while being used for its intended purpose.	☐☐☐☐☐

Open-Ended Question

What suggestions can you give to improve the system?

Answer: ____________________________

C - Map of the Research Locale *(Include Coordinates)*

Latitude: 8° 59' 17.8" N

Longitude: 125° 20' 25.4" E

D - Curriculum Vitae

CURRICULUM VITAE

NAME: Janryl Sacote Signar
ADDRESS: Purok 2, Barangay Matabao, Buenavista Agusan del Norte
EMAIL: janryl_signar@smccnasipit.edu.ph

Personal Information

Sex	: Male
Date of birth	: January 21,2005
Height	: 160cm
Weight	: 53kg
Civil Status	: Single

Educational Attainment

Elementary	: San Francisco Elementary School
Secondary	: Saint James High School
Tertiary	: Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE

NAME: Diether Fedelis Flores
ADDRESS: Purok 7, Barangay Alubihid, Buenavista Agusan del Norte
EMAIL: diether_flores@smccnasipit.edu.ph
Personal Information

Sex	: Male
Date of birth	: October 17 ,2004
Height	: 165cm
Weight	: 54kg
Civil Status	: Single

Educational Attainment

Elementary	: F.s. Omayana Elementary School
Secondary	: Buenavista National High School
Tertiary	: Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE

NAME: Trixia Lee Cabugawan Lope
ADDRESS: D-2 Camagong Nasipit Agusan del Norte
EMAIL: trixialee_lope@smccnasipit.edu.ph

Personal Information

Sex	: Female
Date of birth	: February 23, 2004
Height	: 144cm
Weight	: 41kg
Civil Status	: Single

Educational Attainment

Elementary	: Camagong Elementary School
Secondary	: Saint Michael College of Caraga
Tertiary	: Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE

NAME: Kevin Jun Apos Maestro
ADDRESS: Purok 4, Barangay 1, Buenavista Agusan del Norte
EMAIL: kevinjun_maestro@smccnasipit.edu.ph

Personal Information

Sex	: Male
Date of birth	: November 12, 2004
Height	: 170cm
Weight	: 60kg
Civil Status	: Single

Educational Attainment

Elementary	: Cogon Central Elementary School
Secondary	: Saint James High School
Tertiary	: Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

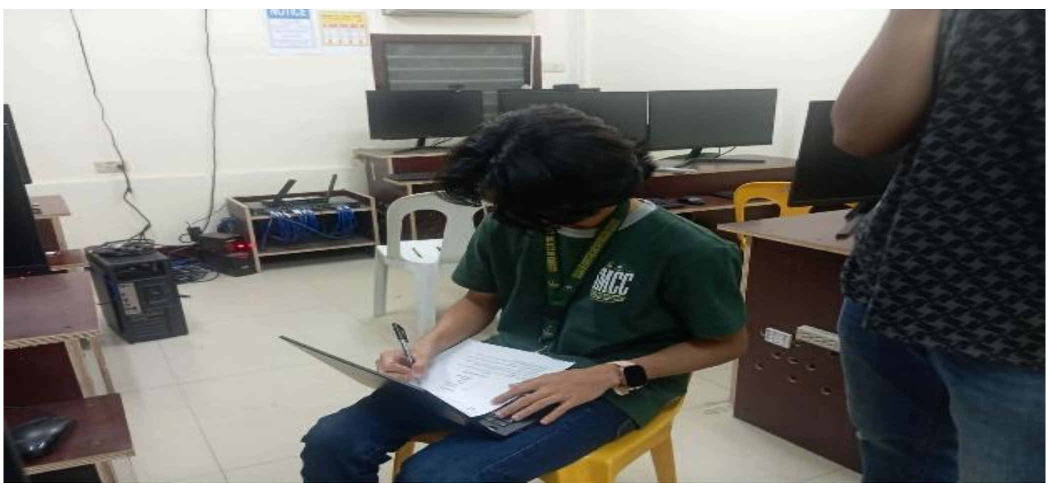
E - Narrative & Photo Documentation

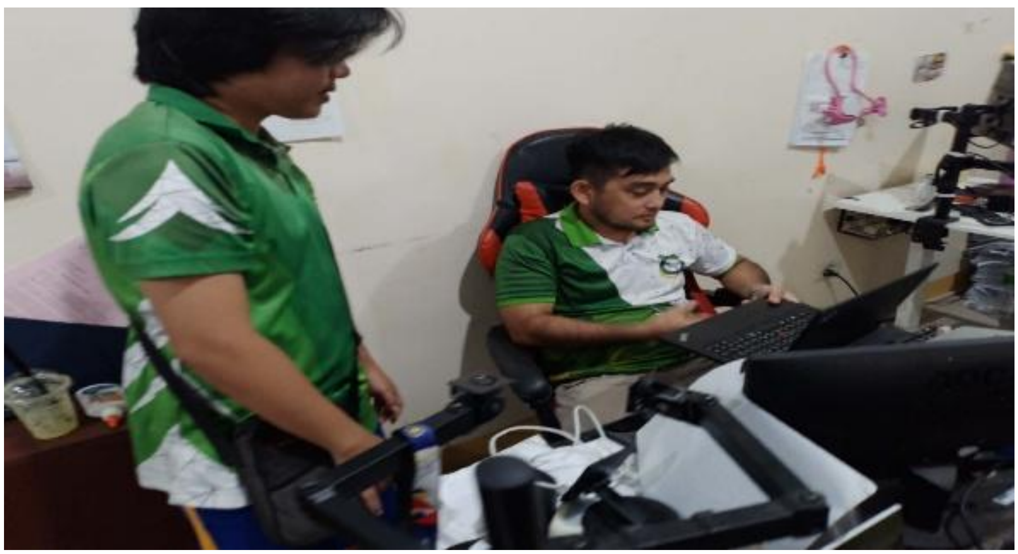
During our Title Proposal

Preparing for Proposal Defense

Final Defense

Evaluation of System Based on ISO 25010





F- Ethics Committee Approval

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

Certificate of Ethics Compliance

This is to certify that the research title **PREDICTING STUDENT ACADEMIC STANDING USING MACHINE LEARINING MODELS WITH EXPLAINABLE AI** with authors ***SIGNAR ET AL.*** passed the research ethics standards and is recommend proceeding with the study.

This certificate is given on the **13th** day of ***May*** **2026** at Saint Michael College of Caraga, Research and Instructional Innovation Department.

MICHELLE ANN G. LUCINO, CREA
Ethics Committee Representative
Saint Michael College of Caraga
Atupan St., Nasipit, Agusan del Norte, Philippines

Members:

Email: communications@smccnasipit.edu.ph

CERTIFICATE OF AUTHENTIC AUTHORSHIP

This is to certify that I/we, the undersigned author(s), have solely and honestly written the entire paper titled: **"Predicting Student Academic Standing Using Machine Learning Models with Explainable AI"** All contents, ideas, data, and analysis presented in this paper are the results of my/our own diligent work and intellectual effort. The paper has been crafted to the best of my/our abilities, in full compliance with academic integrity standards and ethical research practices.

I/We further declare that:

1. This paper is free from any form of plagiarism or copyright infringement.
2. All referenced materials have been properly cited, and due credit has been given to the original authors and sources.
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Signed this
Date:
Location:

Author(s):

Janryl S. Signar
Author

Diether F. Flores
Author

Trixia Lee C. Lope
Author

Kevin Jun A. Maestro
Author